Computational Structural Analysis

전산구조해석

담당교수 : 김 정

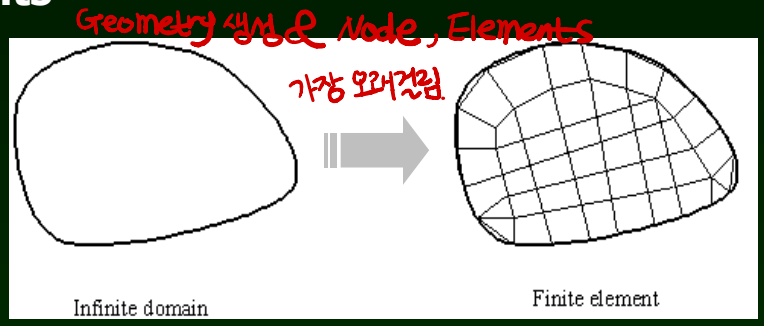
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❑ Basic Steps in the FEM

- **Preprocessing Phase**

(1) create & discretize the solution domain into finite elements

→ nodes and elements



(2) assume a shape function(continuous function) to represent the physical behavior of an element(approximate solution of an element)

(3) develop eqs. for an element (elemental finite element equation)

$$[K]^{(e)} \ , \ \{F\}^{(e)} \longrightarrow [K]^{(e)}\{u\}^{(e)} = \{F\}^{(e)}$$

where, $[K]^{(e)}$: elemental stiffness matrix

$\{u\}^{(e)}$: elemental nodal displacement vector

$\{F\}^{(e)}$: elemental load vector

(4) assemble & construct global stiffness matrix (global equation)

$$[K]^{(G)}\{U\} = \{F\}^{(G)}$$

where, $[K]^{(G)}$: global stiffness matrix

$\{U\}$: global nodal displacement vector

$\{F\}^{(G)}$: global load vector

(5) apply boundary condition, initial condition, loading

- Solution Phase

(6) solve a set of linear or nonlinear algebraic equations

→ primary unknown(displacement or temperature) at each node

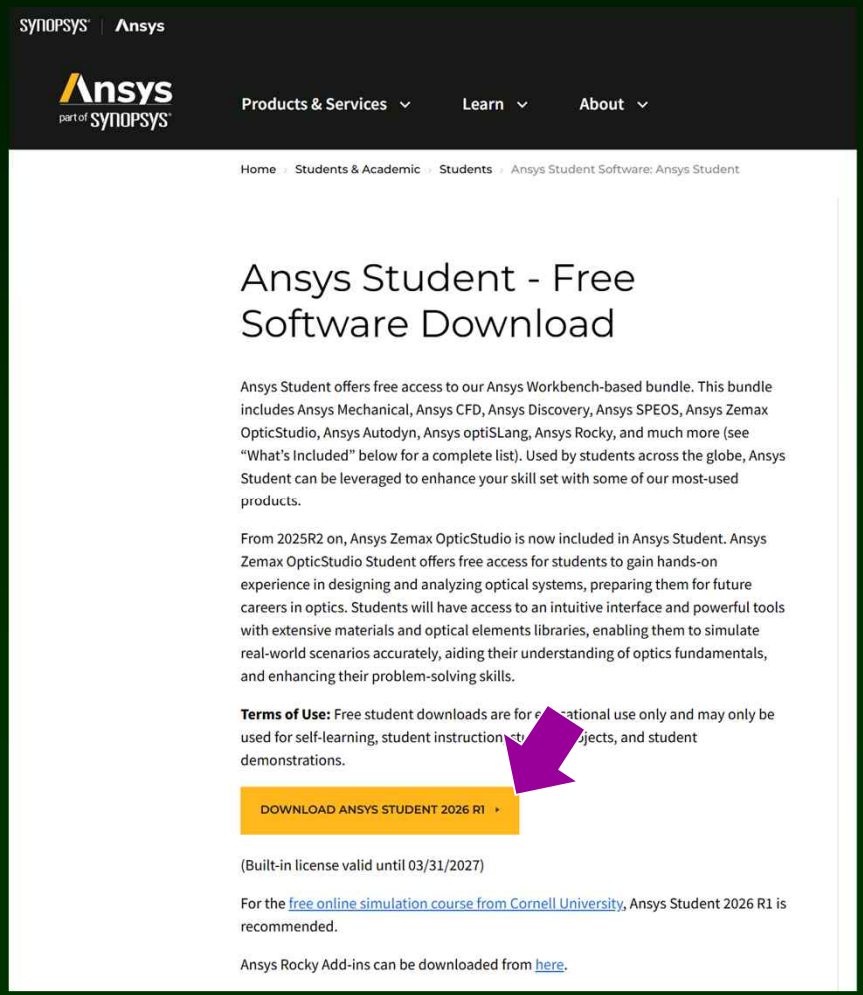
- linear problem : Gaussian Elimination Method is used to solve the considering algebraic equations corresponding to the unknown nodal variables.
- nonlinear problem : some iterative methods such as Newton-Raphson Method, which is based on the Taylor series expansion of the algebraic equations, are used to obtain the final solution.

- Postprocessing Phase

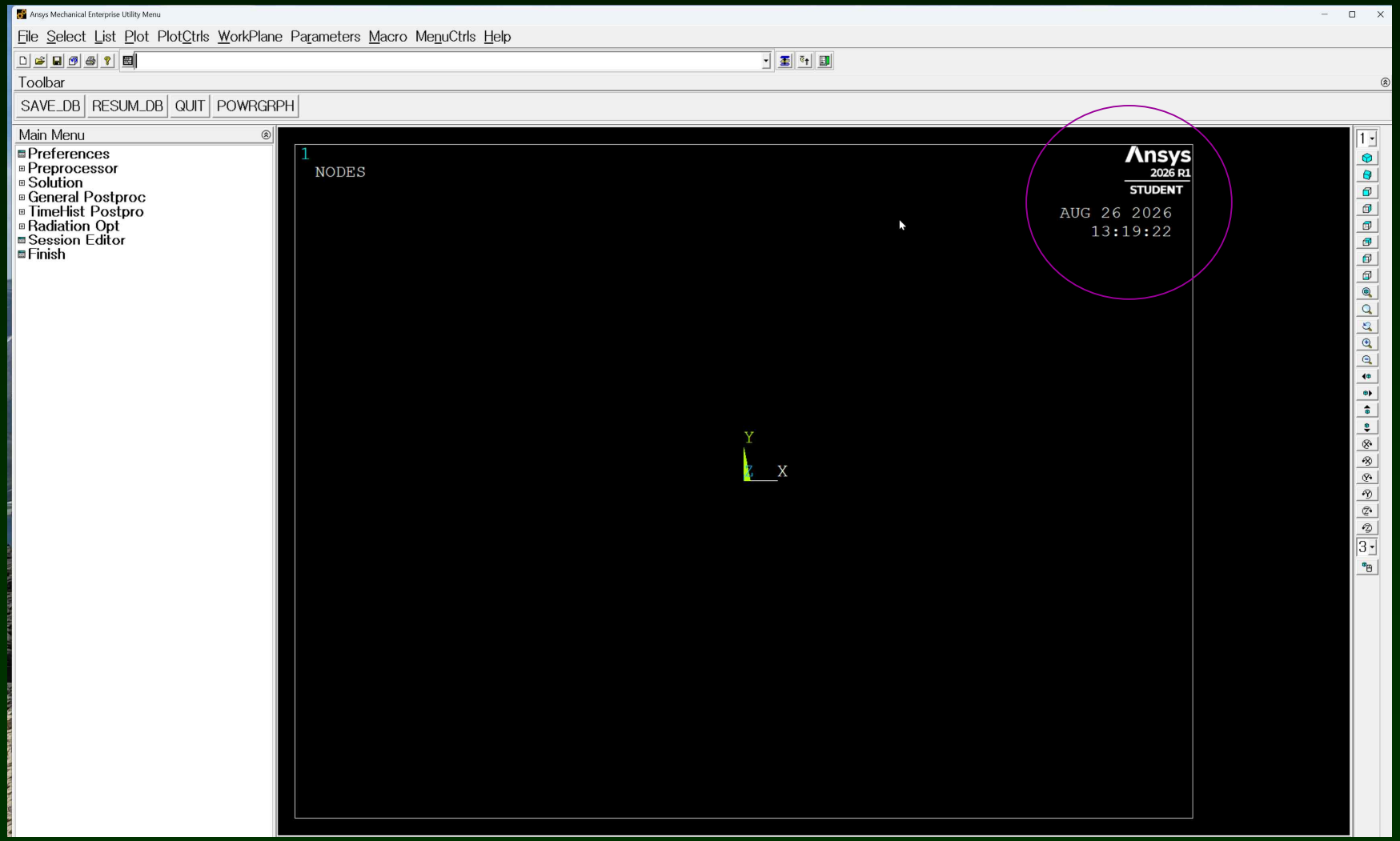
(7) obtain the secondary variables(stress, strain, heat flux etc.)

The ANSYS Program

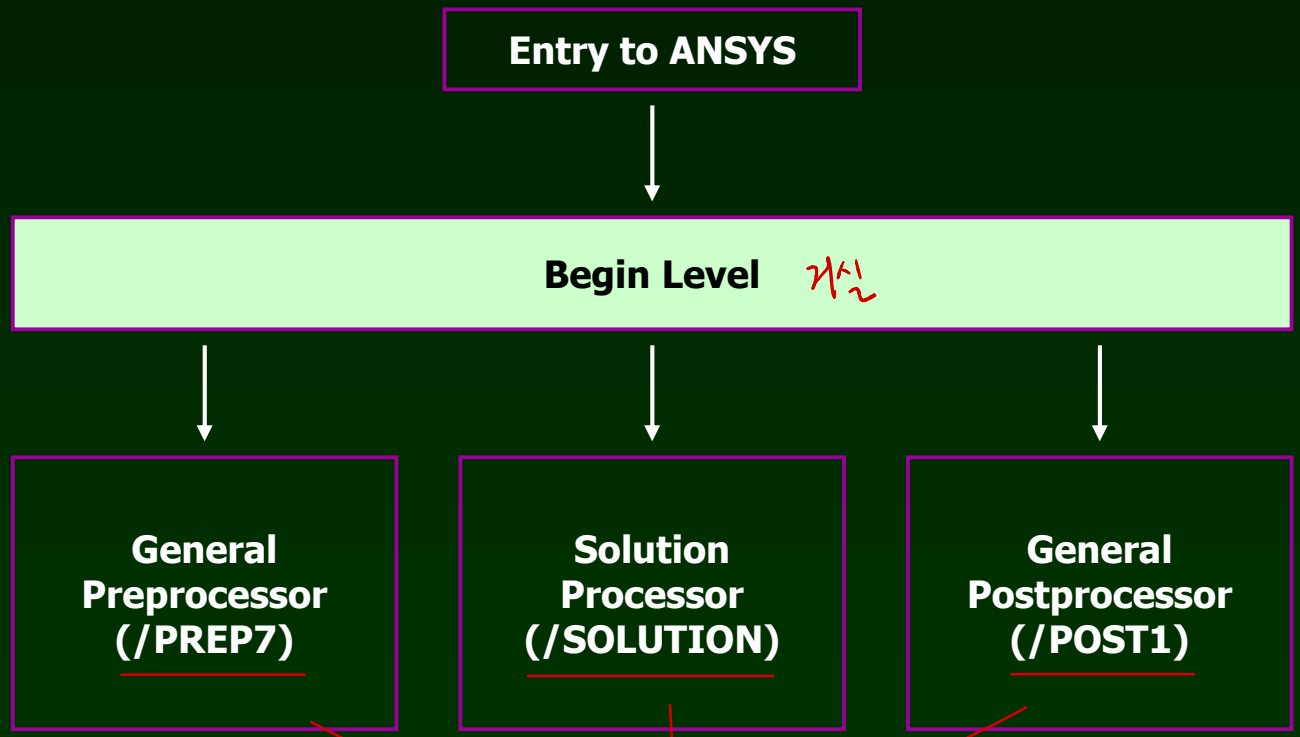
https://www.ansys.com/academic/students/ansys-student



The ANSYS Program



The ANSYS Program



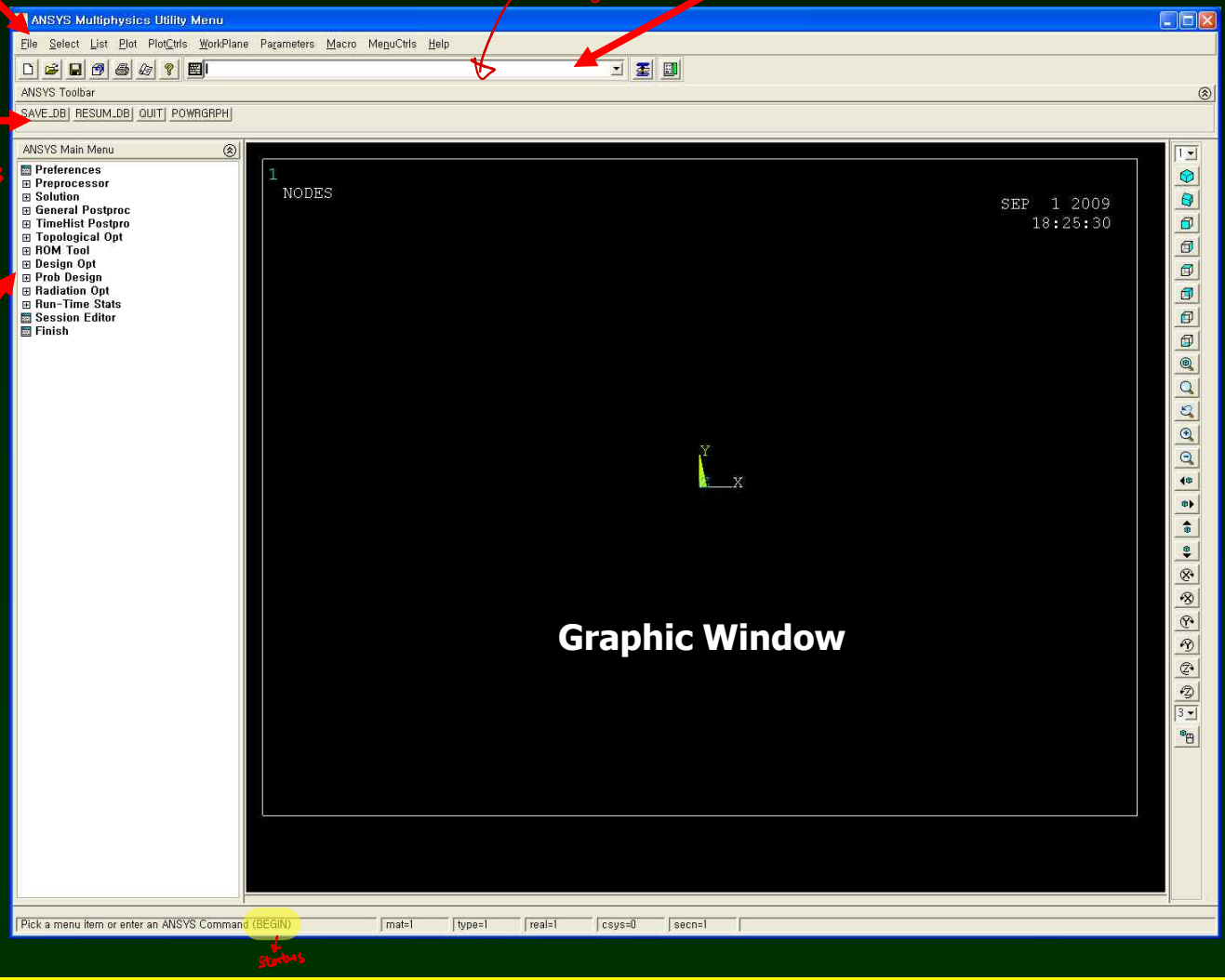
The ANSYS Program

Utility Menu

Input Window

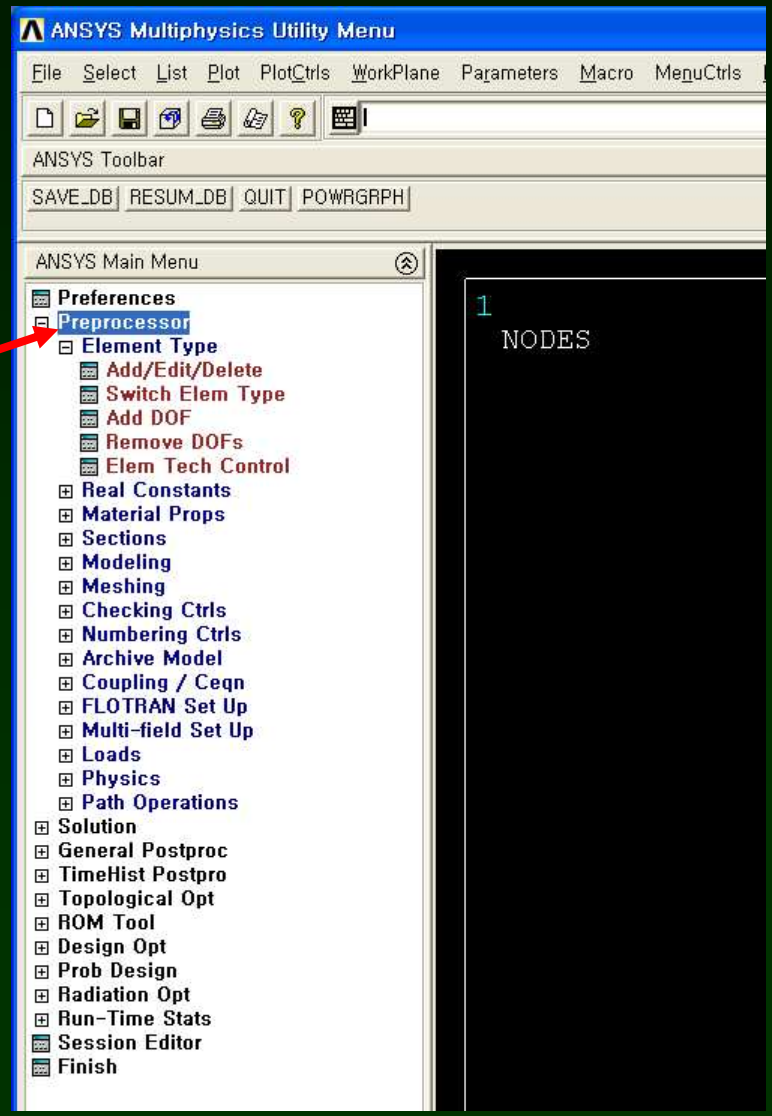
Toolbar

Main Menu



The ANSYS Program

Preprocessor Menu



Some Typical Files

• Log file	File.**LOG**	Coded (ASCII)
• Error file	File.**ERR**	Coded
• Output file	File.**OUT**	Coded
• Database file	File.**DB**	Binary
• Results file	File.**R*xx***	Binary
Structural, Coupled	File.**RST**	
Thermal	File.**RTH**	
Magnetic	File.**RMG**	
• Load step file	File.**S*n***	Coded
• Graphics file	File.**GRPH**	Binary
• Element matrices	File.**EMAT**	Binary

etc.

Modeling – Build the model

- Modeling considerations
- Geometry/Modeling
  - Keypoints
  - Lines
  - Areas
  - Volumes
  - WorkPlane & Coordinate systems
- Element Type : Beam, ...
- Section, Real Constants
- Material Properties
- Meshing

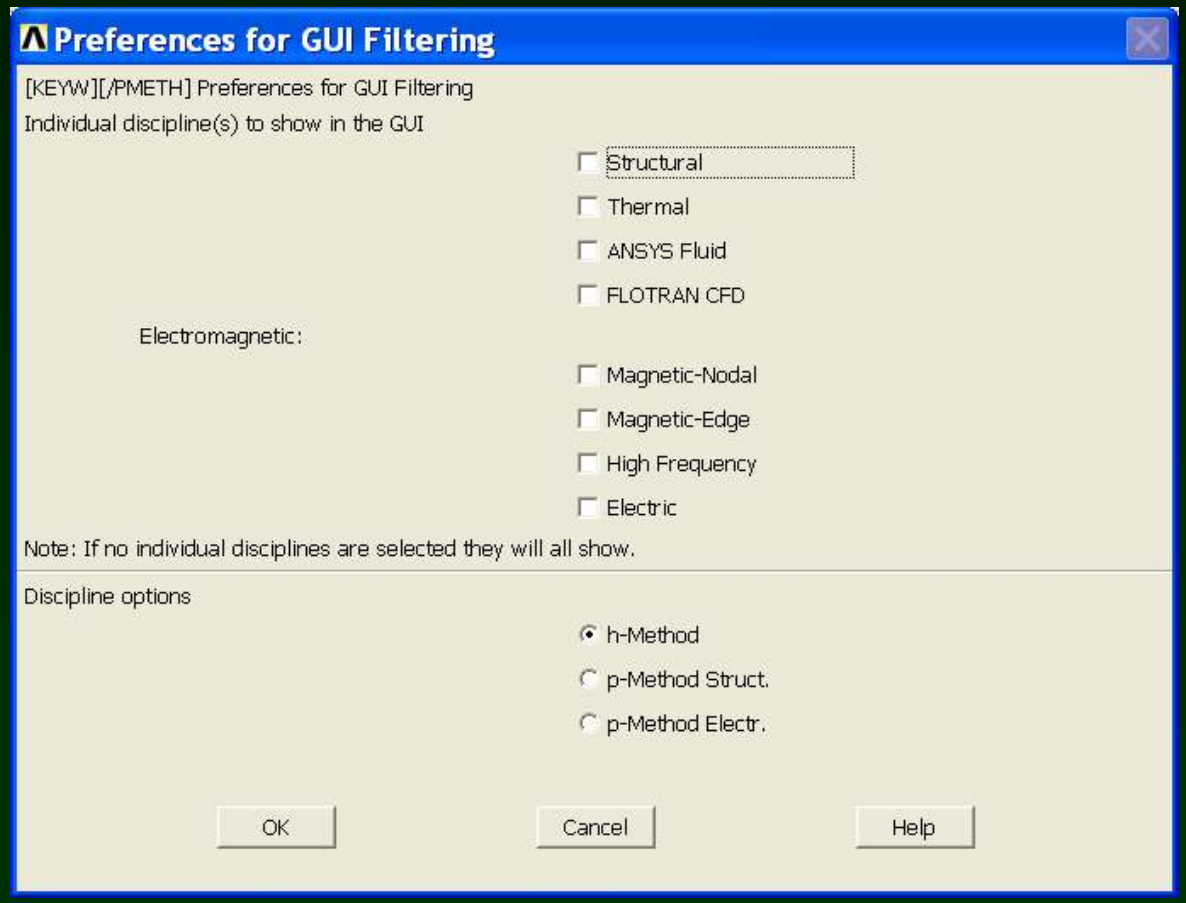
Modeling Considerations

- As you begin your model generation, you will (consciously or unconsciously) make a number of decisions that determine how you will mathematically simulate the physical system:
  - What are the objectives of your analysis?
  - Will you need to vary/modify model data?
  - Will you need to change the geometric topology of the model, e.g. add holes to the model?
  - Will you model all, or just a portion, of the physical system?
  - How much detail will you include in your model?
  - What kinds of elements will you use? How dense should your finite element mesh be?
- In general, you will attempt to balance computational expense (CPU time, etc.) against precision of results as you answer these questions.
- The decisions you make in the planning stage of your analysis will largely govern the success or failure of your analysis efforts.

Modeling Considerations

- Linear or Higher Order Elements
- Take Advantage of Axis-Symmetry
  - The axis of symmetry must coincide with the global Cartesian Y-axis.
  - Negative nodal X-coordinates are not permitted.
  - The global Cartesian Y-direction represents the axial direction, the global Cartesian X-direction represents the radial direction, and the global Cartesian Z-direction corresponds to the circumferential direction.
  - Your model should be assembled using appropriate element types:
    - For axisymmetric models, use applicable 2-D solids with KEYOPT(3) = 1, and/or axisymmetric shells. In addition, various link, contact, combination, and surface elements can be included in a model that also contains axisymmetric solids or shells. (The program will not realize that these "other" elements are axisymmetric unless axisymmetric solids or shells are present.)
- How Much Detail to Include
- Appropriate Mesh Density

Modeling Considerations



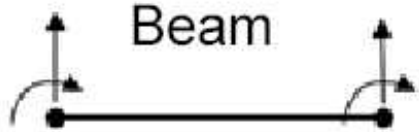
Modeling Considerations

- **Characterization of problem**

Rod



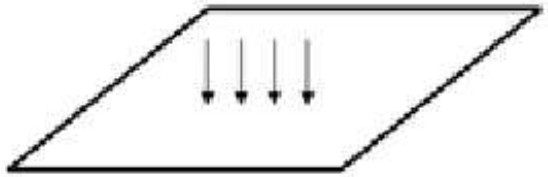
Beam



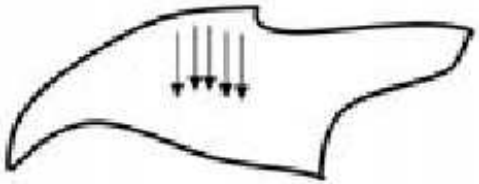
Disk



Plate



Shell



Solid



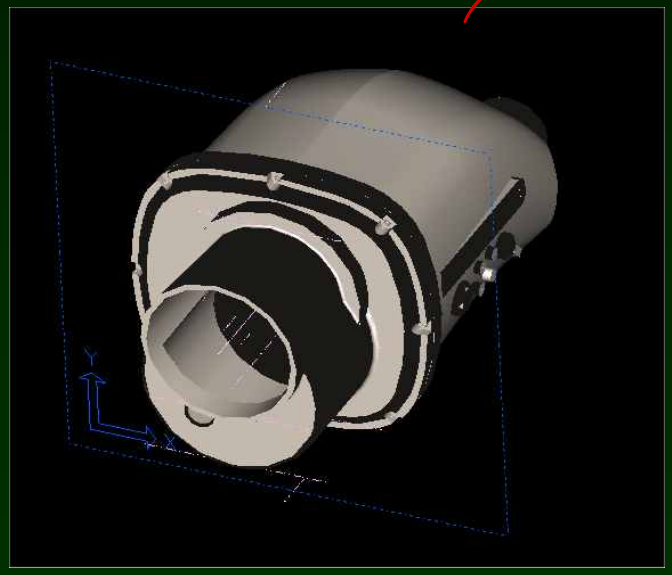
Modeling Considerations

- The ANSYS program does not assume a system of units for your analysis.
- Units must however be consistent for all input data.
- Geometry/Modeling
  - Creating a solid model within ANSYS.
  - Using direct generation.
  - Importing a model created in a computer aided design (CAD) system. (e.g. IGES, Catia, Creo Parametric, NX, ACIS, Parasolid)

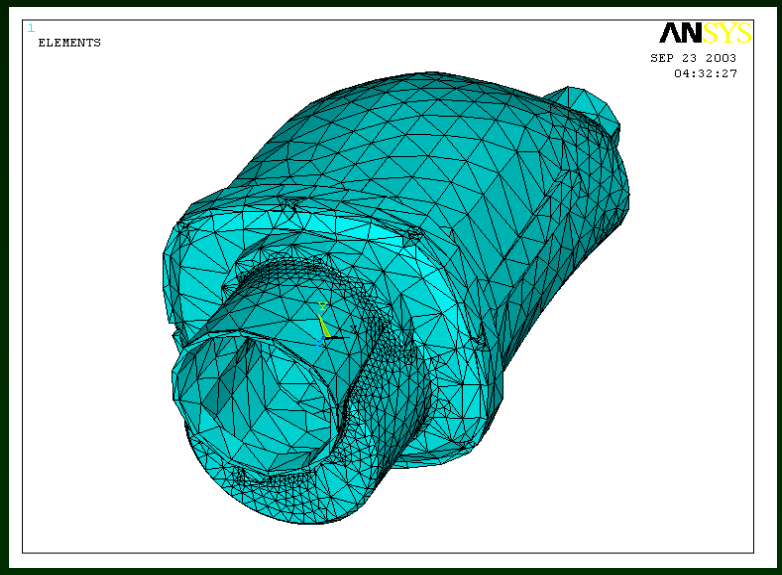
Modeling Considerations

- **Discretize the solution domain into finite elements**

Solid Modeling



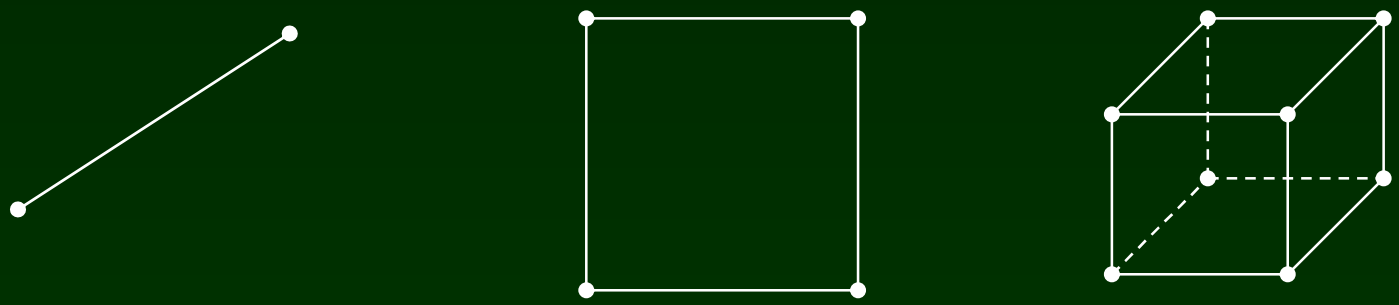
Engineering System



Possible Finite Element Model

- **Node(절점)** : 공간에서의 좌표로 표시되며, 여기서 자유도(변위, 온도, 場 등), 상태(힘, 전류, 열류 등)가 계산된다.

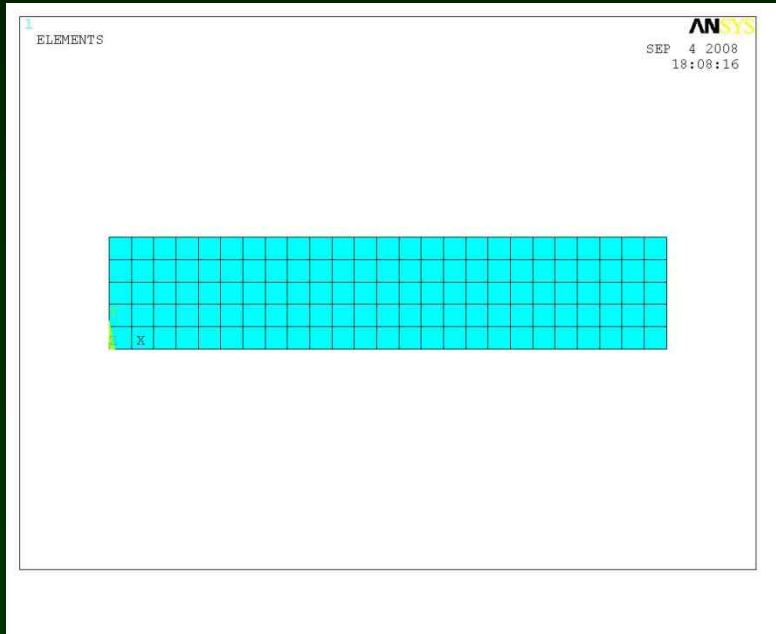
- **Element(요소)** : 여러 개의 절점으로 구성되며, 절점들의 자유도에 의거 수학적 해를 위한 행렬 등이 구성된다. 요소에는 선, 평면, 혹은 부피를 가진 **Solid** 등이 있다.

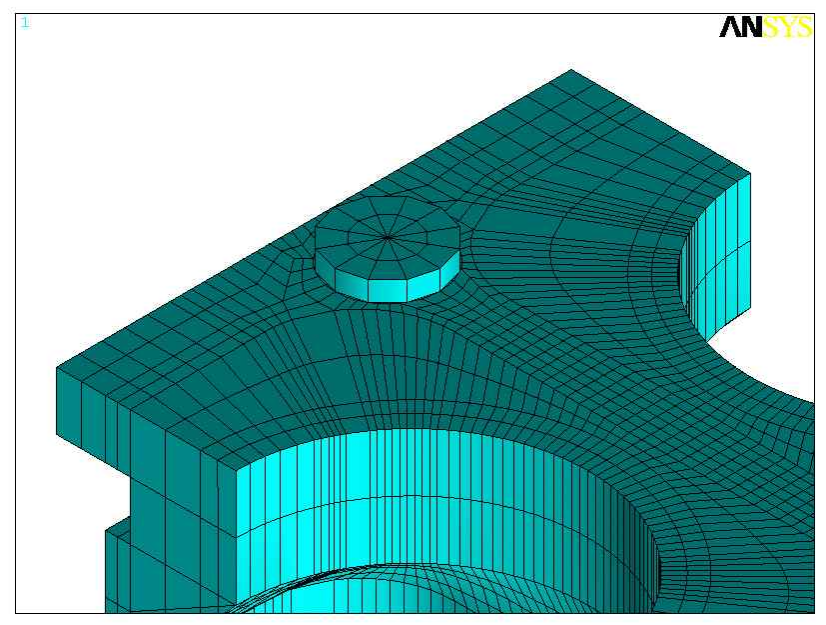


1D **2D** **3D**

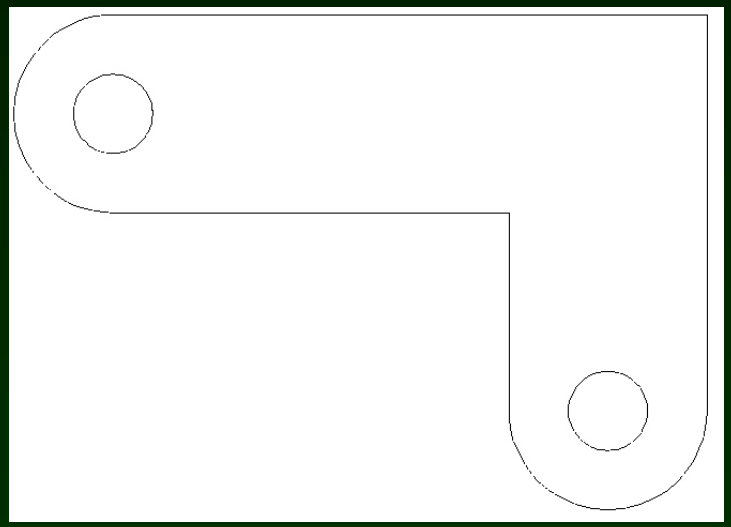
❑ **Finite Element Model**을 구축하는 **2**가지 방법

- **Solid Modeling** → ANSYS own Geometry 만들어야함. (CAD works), Node & Element Automatically Generated.
- **Direct Generation** → Geometry 만들지 않고, Node & Element 직접, Manually



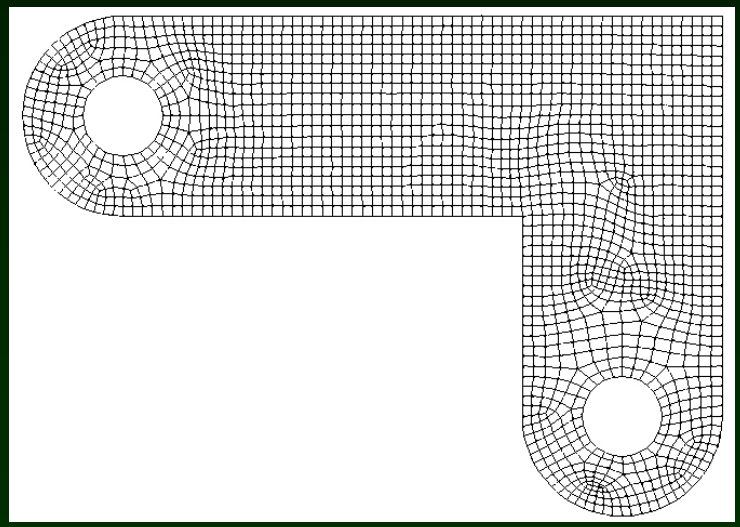


- **Solid Modeling :** 복잡한 형상의 **Model** 구축시 강력하고 다양한 기법을 제공
- **Meshing : Solid Model**내에 **Node**와 **Element** 등을 생성시키는 작업



Solid Model

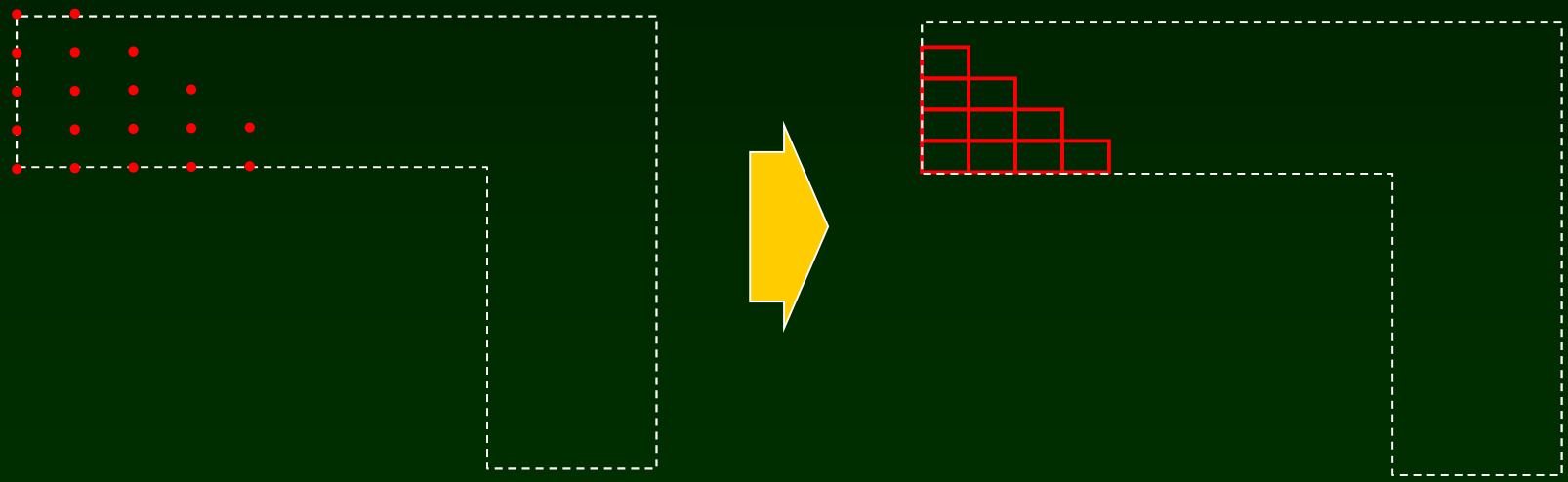
Model represented by geometric shapes (rectangles, circles, polygons, …, other arbitrary shapes)



Finite Element Model

Nodes and elements automatically generated by the program based on user-specified element size and shape

- **Direct Generation :** 사각형 같은 단순한 형상을 **Meshing**할 때 사용되며, 간단하면서도 신속함
- 수작업으로 **Node**와 **Element** 생성

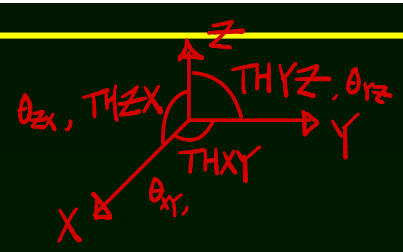


Nodes

Nodes are "manually" specified at desired locations

Elements

Element connectivities are then established, also manually



TH: Theta (θ)

Direct Generation : 직접 NODE, ELEMENT 생성

- Direct Generation 방법은 User가 Node와 Element를 직접 생성한다.

※Begin창 아님, PREP7창.

- Define a node

N, *Node* (zero or negative), *X*, *Y*, *Z*, *THXY*, *THYZ*, *THZX*

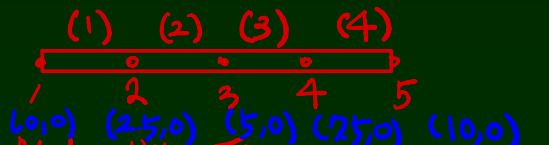
Node number

Node location and rotation

location / Rotation

Pre processing 에서 command "N" Begin X

ex) N,2,2.5,0,0
N,3,5,0,0
} ,0은 생략 Default : 0



Node 번호 필수 → 좌표값 필수

L=10 이면 간격은 2.5

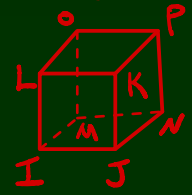
- Define a element by node connectivity

E, *I*, *J*, *K*, *L*, *M*, *N*, *O*, *P*

1 2 3 4 5 6 7 8

Node numbers assigned to node *I* ~ node *P*

3D가 가장 많은 nodes 필요



ET : Element Type
EN : Element Node

Element 1) node 1 & 2 : E,1,2 or EN,1,1,2
" 2) node 2 & 3 : E,2,3 or EN,2,2,3

Element Number 생략 가능

Bottom-up VS TOP-down
· Point → line → Area → Volume
· Volume 만듦
· 나머지 형상은 자동지정
· Primitive 형상 → 가공해서 desired shape

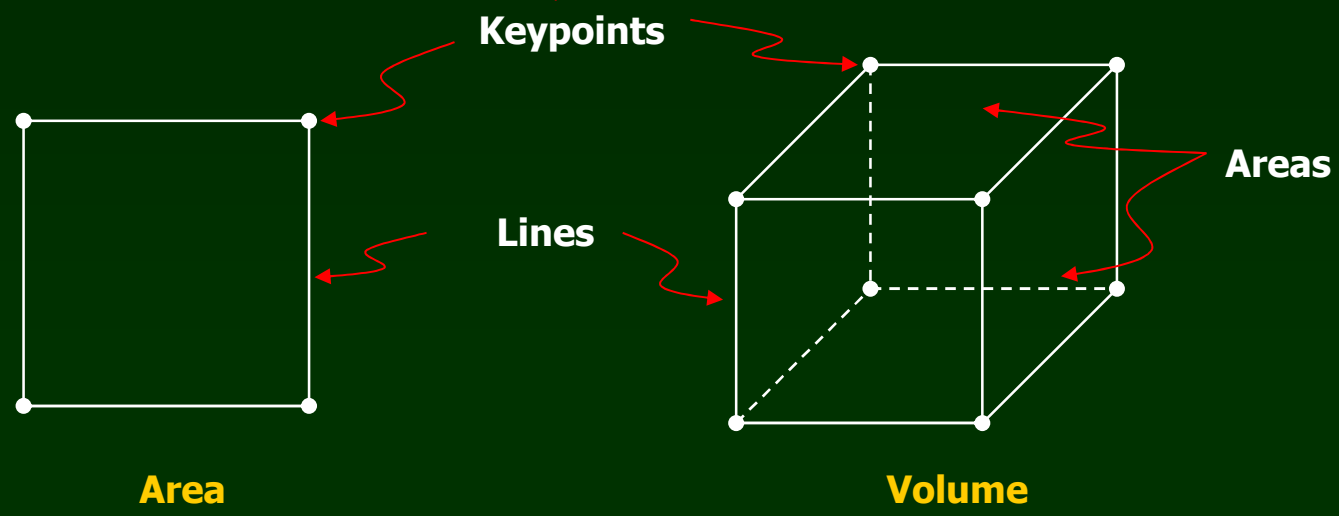
Solid Modeling → CAD 작업. line, Area, Volume... Node, Element는 ANSYS 자동생성.

Creating Your Solid Model from the Bottom Up

점 8개에서 시작

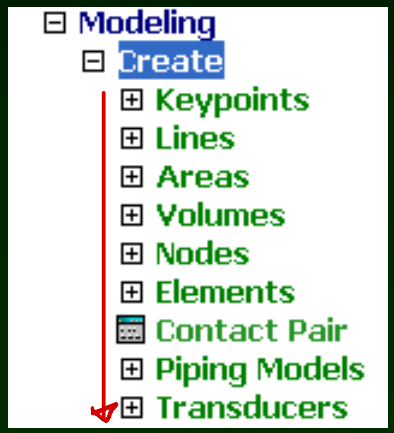
- Keypoints are the vertices, lines are the edges, areas are the faces, and volumes are the interior of the object.
- Volumes, the highest-order entities, are bounded by areas, which are bounded by lines, which in turn are bounded by keypoints.

→ 3D에서 주요 점. (X, Y, Z)

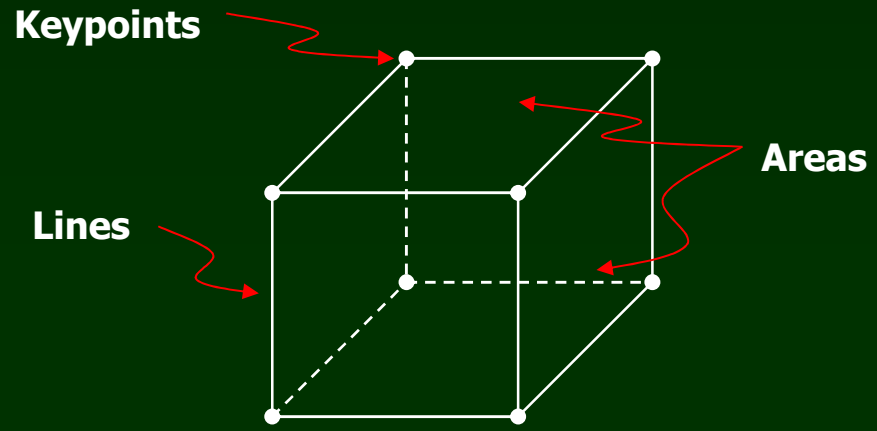


Solid Modeling

Creating Your Solid Model from the Bottom Up



- The hierarchy of modeling entities is as listed below:
  - Elements (and Element Loads)
  - Nodes (and Nodal Loads)
  - Volumes (and Solid-Model Body Loads)
  - Areas (and Solid-Model Surface Loads)
  - Lines (and Solid-Model Line Loads)
  - Keypoints (and Solid-Model Point Loads)



Solid Modeling

Creating Your Solid Model from the Bottom Up

- Define a keypoint

  K, *NPT*, *X*, *Y*, *Z*

- Define a line

  L, *P1*, *P2*, *NDIV*, *SPACE*, *XV1*, *YV1*, *ZV1*, *XV2*, *YV2*, *ZV2*

- Define an area

  A, *P1*, *P2*, *P3*, *P4*, *P5*, *P6*, *P7*, *P8*, *P9*, *P10*, *P11*, *P12*, *P13*, *P14*, *P15*, *P16*, *P17*, *P18*

  AL, *L1*, *L2*, *L3*, *L4*, *L5*, *L6*, *L7*, *L8*, *L9*, *L10*

- Define a volume

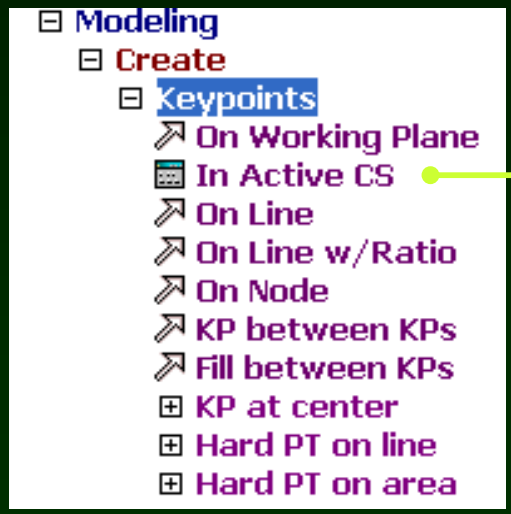
  V, *P1*, *P2*, *P3*, *P4*, *P5*, *P6*, *P7*, *P8*

  VA, *A1*, *A2*, *A3*, *A4*, *A5*, *A6*, *A7*, *A8*, *A9*, *A10*

Solid Modeling

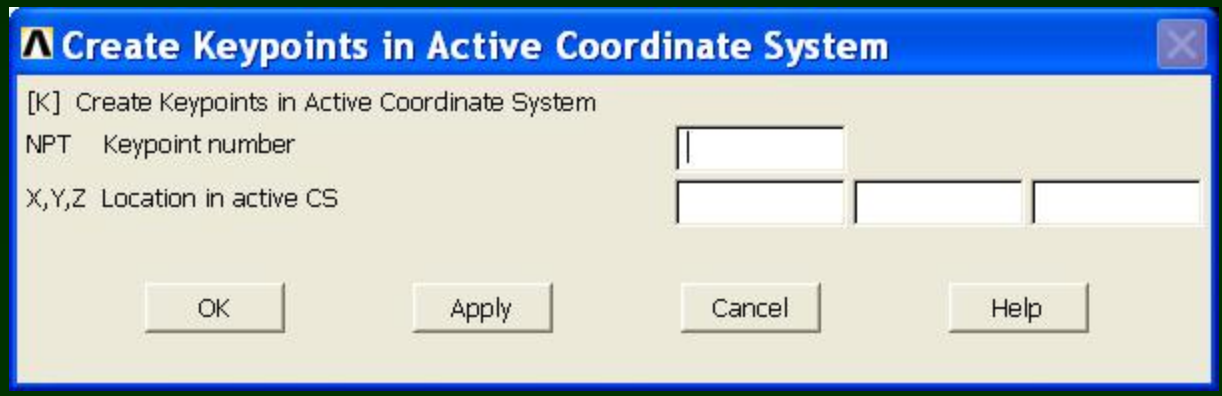
Creating Your Solid Model from the Bottom Up

- Define a keypoint



〈Manual, GUI 방식, 추천 X〉

It is a good idea to use keypoints as reference points in the modeling phase



Solid Modeling

Creating Your Solid Model from the Top Down

- Creating your model using geometric primitives, which are fully-defined lines, areas, and volumes.
- As you create a primitive, the program automatically creates all the "lower" entities associated with it.
- You can freely combine bottom up and top down modeling techniques, as appropriate.

2-D Primitives

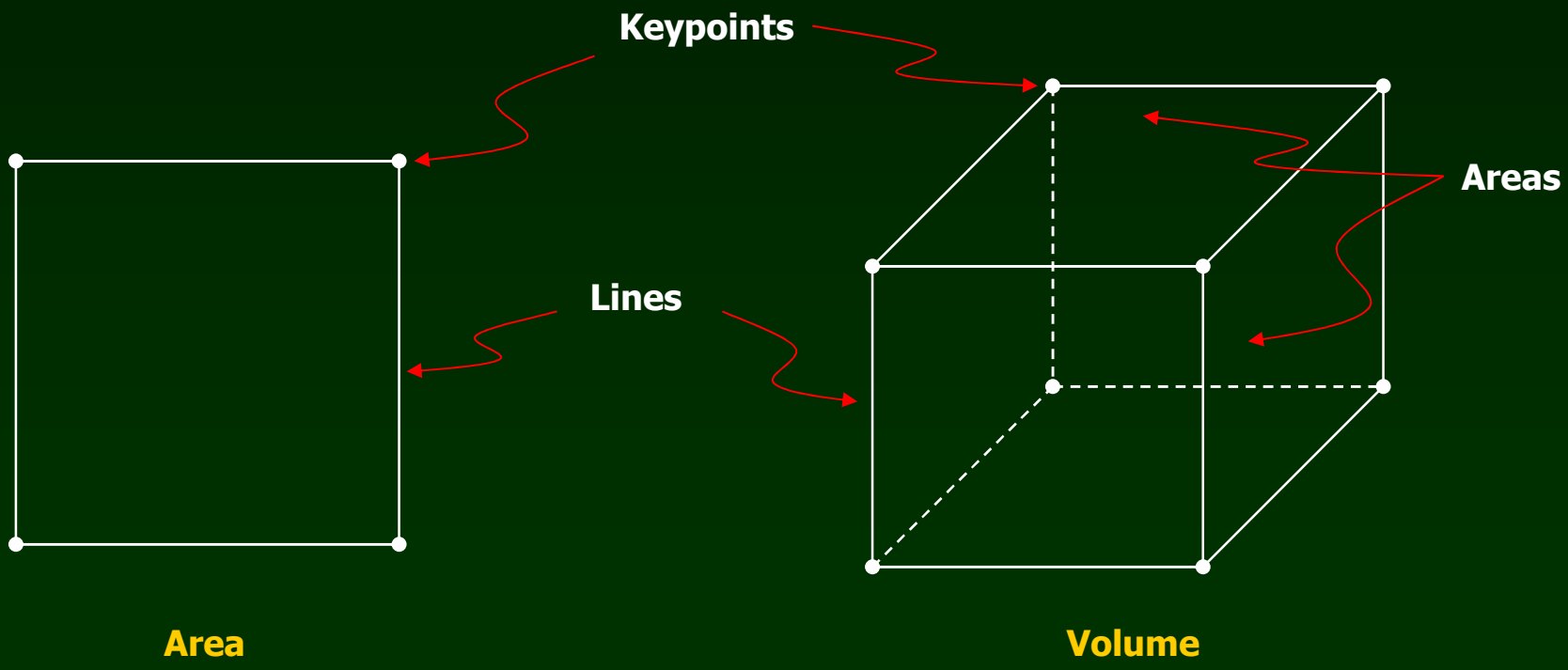
Rectangle
Circle
Polygon

3-D Primitives

Block
Prism
Cylinder
Cone
Sphere
Torus

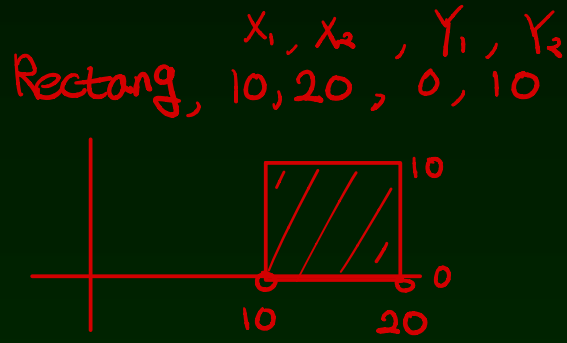
- **2-D**와 **3-D**의 **Primitive**는 "**Area**"와 "**Volume**"으로 각각 불리어진다.
  Area와 **Volume**은 **program**내에서 번호로 구분된다.

- **Volume**은 **Area**로 구축, 구분되며 **Area**는 선**(Line)**, 선은 점**(Keypoint)**으로 구축, 구분된다.

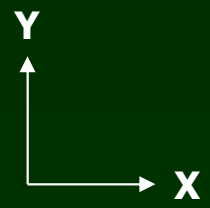


• **Rectangular**는 아래의 **Command**로 **4**개의 좌표를 지정하여 생성한다.

RECTANG, *X1*, *X2*, *Y1*, *Y2*





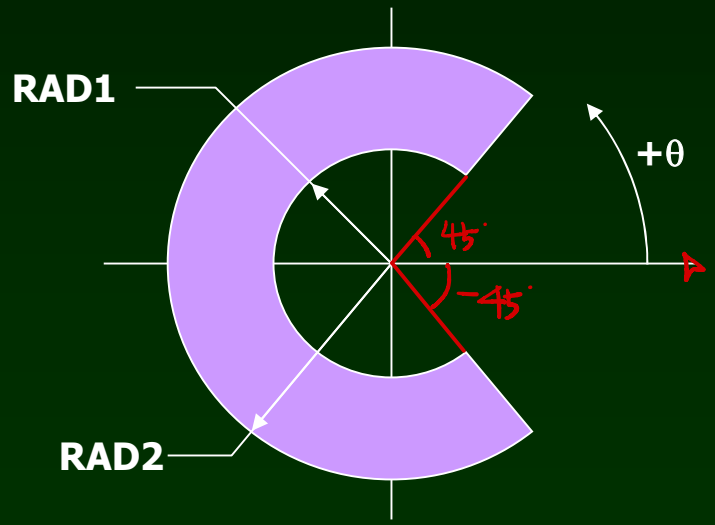


- 도우넛 모양**(Circular area)** 생성 방법

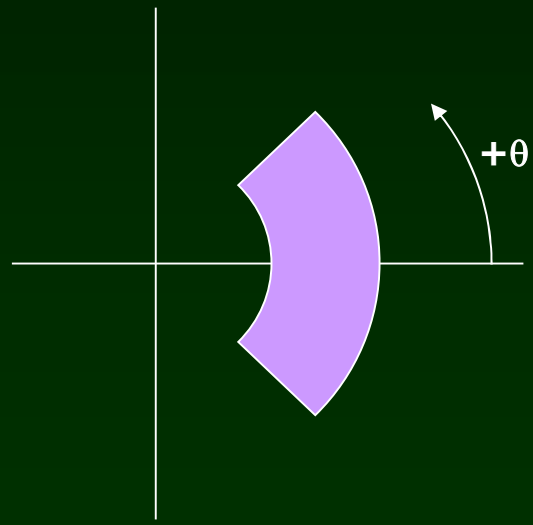
Full circle
PCIRCLE (5, 0, 360)

Inner radius '작은값'
Polycircle ↑ ↗ Outer radius '큰값'

PCIRCLE, *RAD1*, *RAD2*, *THETA1*, *THETA2*



PCIRC, 1, 2, 45, 315
PCIRC, 1, 2, 315, 45



PCIRC, 1, 2, -45, 45
PCIRC, 1, 2, 45, -45

- Graphic display commands

Volume
VPLOT to display volumes

Area
APLOT to display areas

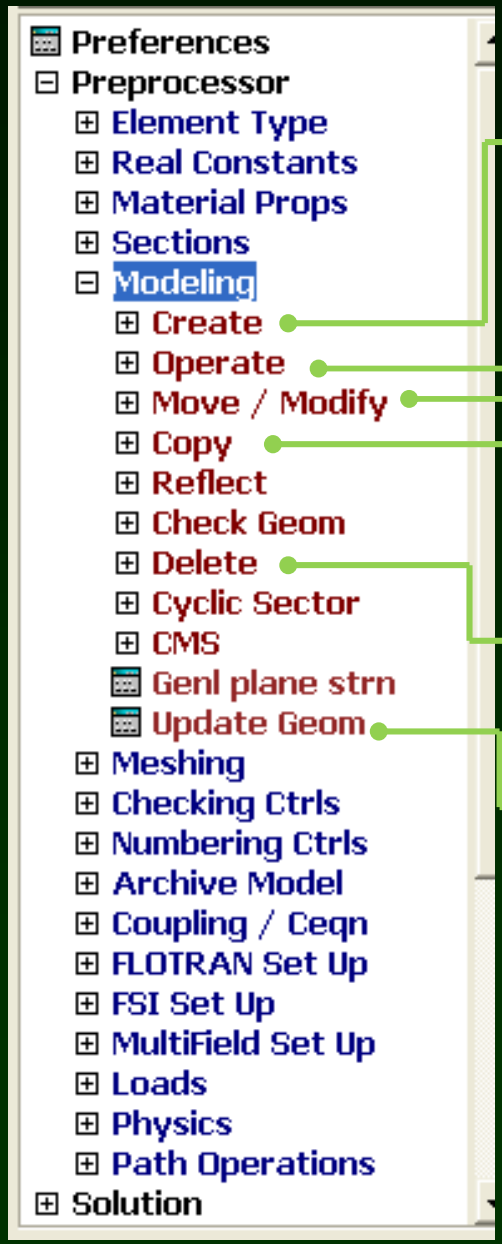
Line
LPLOT to display lines

KeyPt
KPLOT to display keypoints

NODE
NPLOT to display nodes

Element
EPLOT to display elemens

NO "UNDO" → Delete & Modify



Create – geometrical entities

Operate – perform Boolean operations

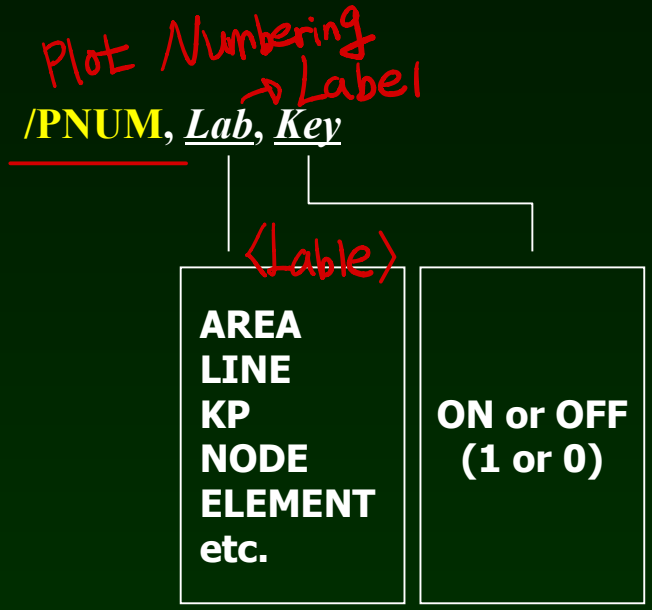
Move / Modify – move or modify geometrical entities

Copy – copy geometrical entities

Delete – delete geometrical entities

Update Geom – update the geometry in relation to for example buckling analysis

- **Display numbering command**

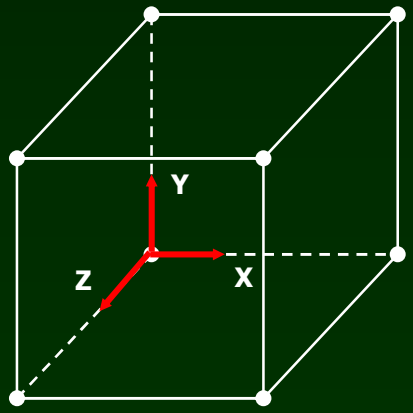


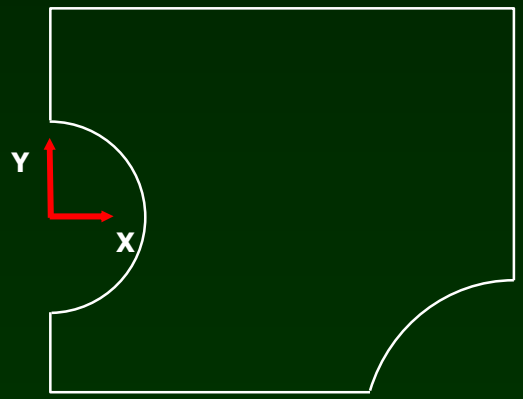
/PNUM is just a setting or specification. Issuing the command does not instantly put numbers on your screen. You need to follow it with an action command such as APLOT (or LPLOT, KPLOT, etc.) for the specification to take effect.

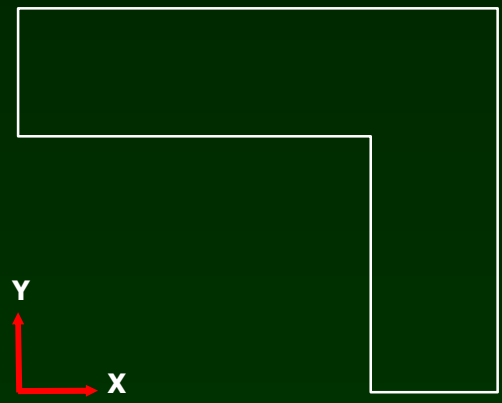
Global Coordinate System

- 모든 **Model**은 그 크기를 표시하기 위해서 기준이 되는 점이 있어야 하는데, 이를 **Global origin**이라 한다.
- **Global origin**은 편리에 따라 어떤 임의의 점으로도 선정 가능하다.

* 수정불가.







Types of Coordinate System

The ANSYS program has several types of coordinate systems, each used for a different reason:

수정가능, 여러개 가능.

•Global and local coordinate systems are used to locate geometry items (nodes, keypoints, etc.) in space.

•Display coordinate system determines the system in which geometry items are listed or displayed.

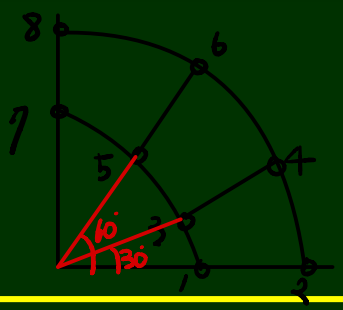
•Nodal coordinate system defines the degree of freedom directions at each node and the orientation of nodal results data.

•Element coordinate system determines the orientation of material properties and element results data.

•Results coordinate system is used to transform nodal or element results data to a particular coordinate system for listings, displays, or general postprocessing operations (POST1).

Global Coordinate System

- A global coordinate system can be thought of as an absolute reference frame.
- The ANSYS program provides three predefined global systems: Cartesian, cylindrical, and spherical.
- All three of these systems are right-handed and, by definition, share the same origin. They are identified by their coordinate system (C.S) numbers: 0 for Cartesian, 1 for cylindrical, and 2 for spherical.
  - **Cartesian** (X, Y, Z components) coordinate system 0 (KCN, KCS=0)
  - **Cylindrical** (R, θ, Z components) coordinate system 1 (KCN, KCS=1)
  - **Spherical** (R, θ, φ components) coordinate system 2 (KCN, KCS=2)




```
/PREP
CSYS, 1 : Cylindrical
N,1,1
N,2,2
N,3,1,30
N,4,2,30
N,5,1,60
N,6,2,60
N,7,1,90
N,8,2,90
```

& nlist (node list) & nplot

/PSYMB
: Symbol Plot

Local Coordinate System

- In many cases, it may be necessary to establish your own coordinate system, whose origin is offset from the global origin, or whose orientation differs from that of the predefined global systems.
- Such user defined coordinate systems, known as local coordinate systems, can be created in the following ways:
  - Define the local system in terms of global Cartesian coordinates.
  - Command(s): LOCAL
    GUI: Utility Menu> WorkPlane> Local Coordinate Systems> Create Local CS> At Specified Loc

local CS의 번호지정 (10번이상 ※) Coordinate System (0, 1, 2)

LOCAL, *KCN, KCS, XC, YC, ZC, THXY, THYZ, THZX, PAR1, PAR2*

Reference number

원점위치 (Origin of local coor.)

Rotation

local, 10, 1 : Cylindrical, 10번, local Coordinate

6번 node의 생성 : local, 11, 1, 2*cos(60x3.14/180), 2*sin(")

Global coordinate 기준 node 6 위치, (Radian)

Active Coordinate System

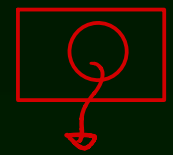
LSYS, 1

You may define as many coordinate systems as you like, but only one of these systems may be active at a time. The choice of active coordinate system is determined as follows:

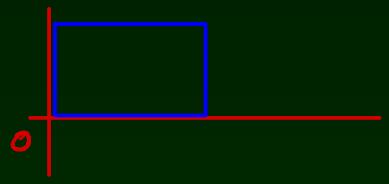
- Initially, the global Cartesian system is active by default.
- Each time you define a local coordinate system, that newly-defined system then automatically becomes the active one.
- If you want to activate one of the global coordinate systems or some other previously defined coordinate system, use one of the following methods:
  - Activates a previously defined coordinate system.
  - Command(s): CSYS , 이미 있는 coordinate system을 activation 시키는 command.
    GUI: Utility Menu> WorkPlane> Change Active CS to> Global Cartesian
    Utility Menu> WorkPlane> Change Active CS to> Global Cylindrical
    Utility Menu> WorkPlane> Change Active CS to> Global Spherical
    Utility Menu> WorkPlane> Change Active CS to> Specified Coord Sys
    Utility Menu> WorkPlane> Change Active CS to> Working Plane

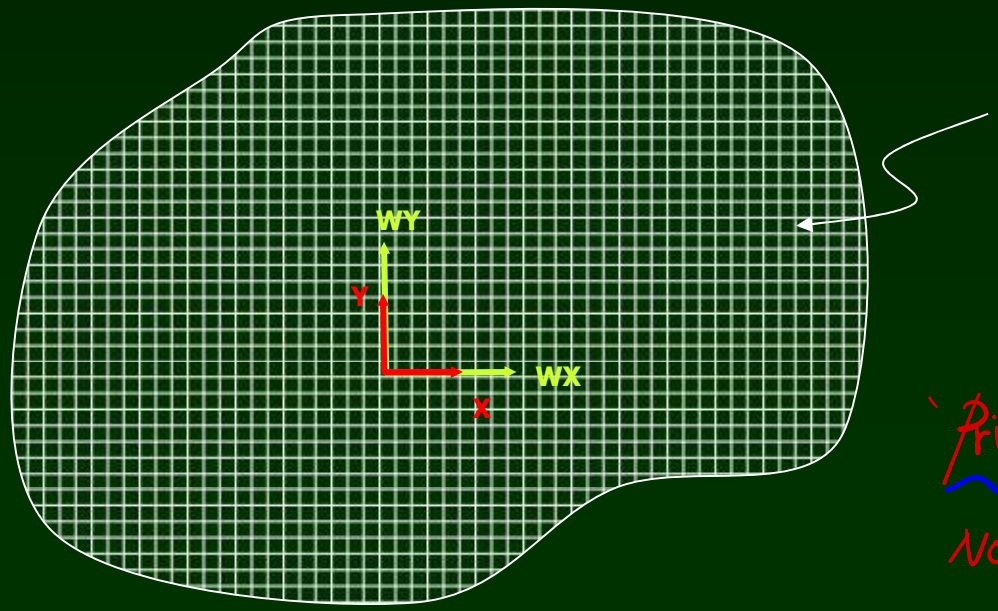
CSYS, *KCN*

Working Plane (WP) : 작업 평면

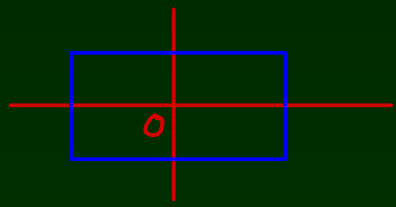


- **Primitive**의 위치, 방향이 정해지는 기준이 되는 평면
- "**Working plane coordinate system**"이라는 이미 만들어진 **2-D** 직교 좌표계가 있다.
  Default로 이는 **Global coordinate system**과 일치한다.





Working plane



- **Moving and Rotating the Working Plane**

WPAVE, *X1, Y1, Z1*

Coordinate (in the currently active global or local coordinate system) of the new WP origin (*X1, Y1, Z1*)

WPOFFS, *XOFF, YOFF, ZOFF*

Coordinate (in the currently active working plane coordinate system) of the new WP origin (*XOFF, YOFF, ZOFF*)

WPROTA, *THXY, THYZ, THZX*

Rotation angels (*THXY, THYZ, THZX*)

WPSTYL, *DEFA*

AND, OR, (true or false)

Boolean Operations

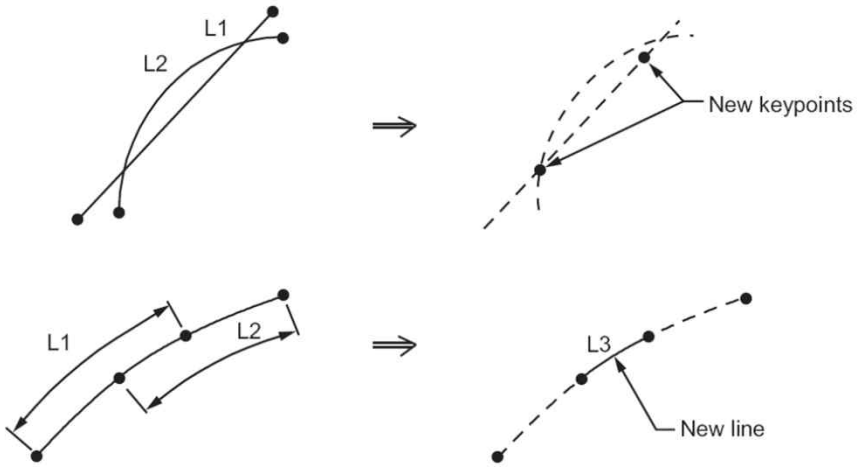
- Intersect
- Add
- Subtract
- Classify
- Overlap
- Glue
- Partition

Booleans - Intersect

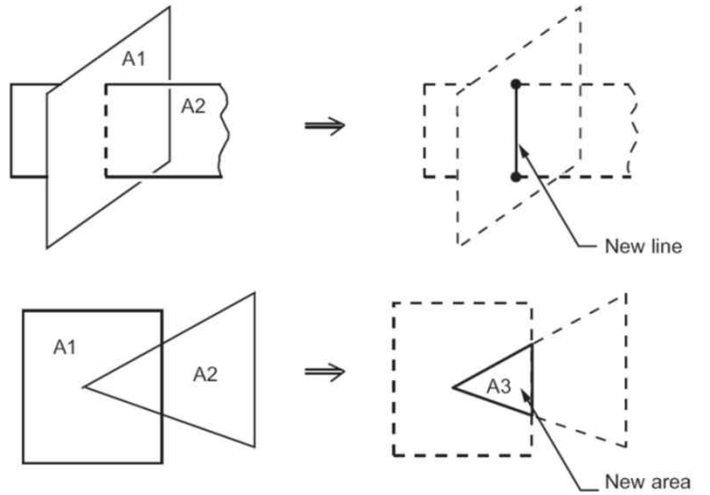
□IN△ : □이 △와 Intersect

* 다른 차원끼리 Intersect 가능.

LINL (Line Intersect Line)



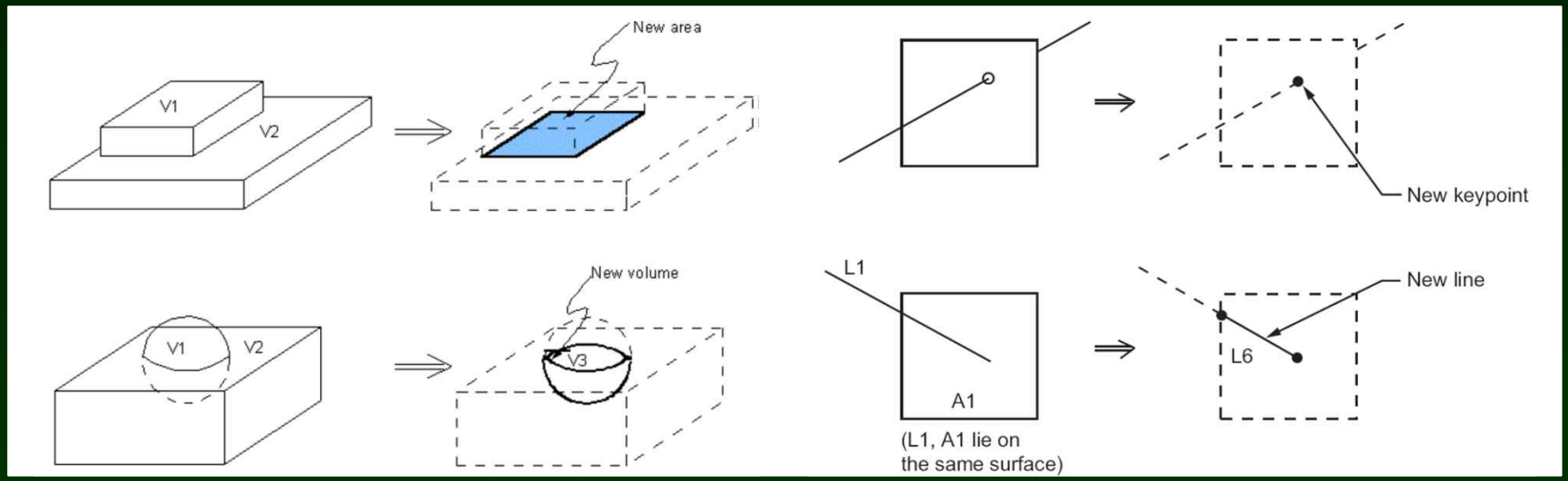
AINA (Area Intersect Area)



Booleans - Intersect

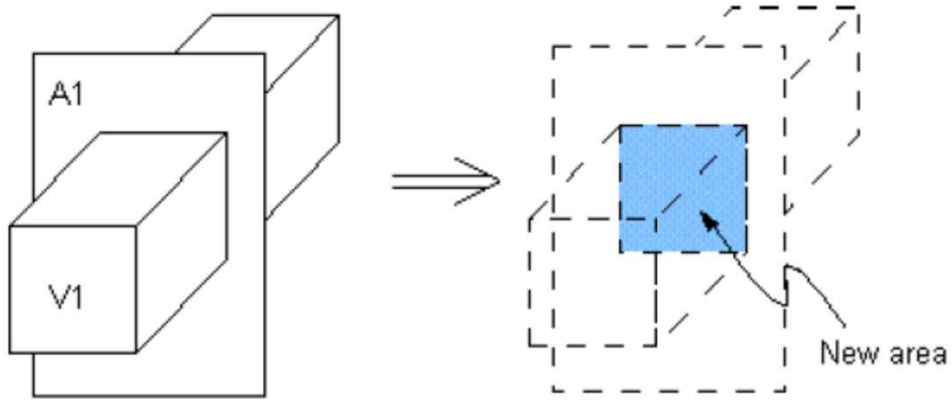
VINV (Volume Intersect Volume)

LINA (Line Intersect Area)

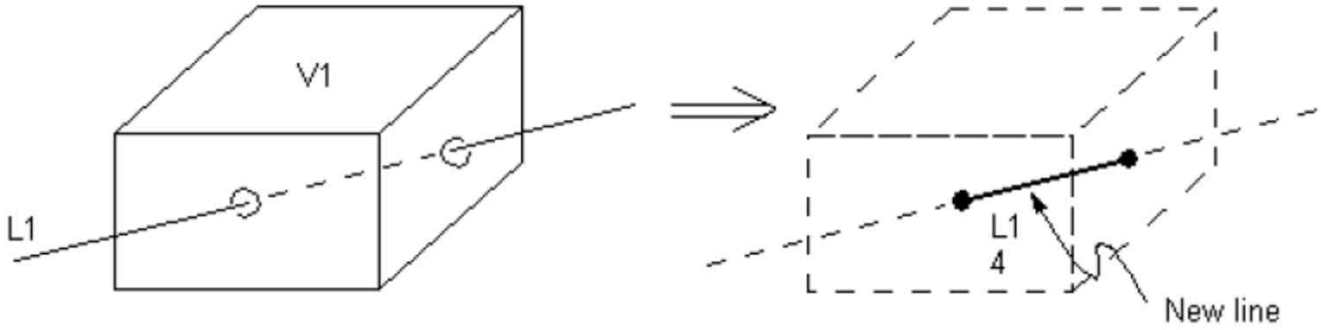


Booleans - Intersect

AINV (Area Intersect Volume)

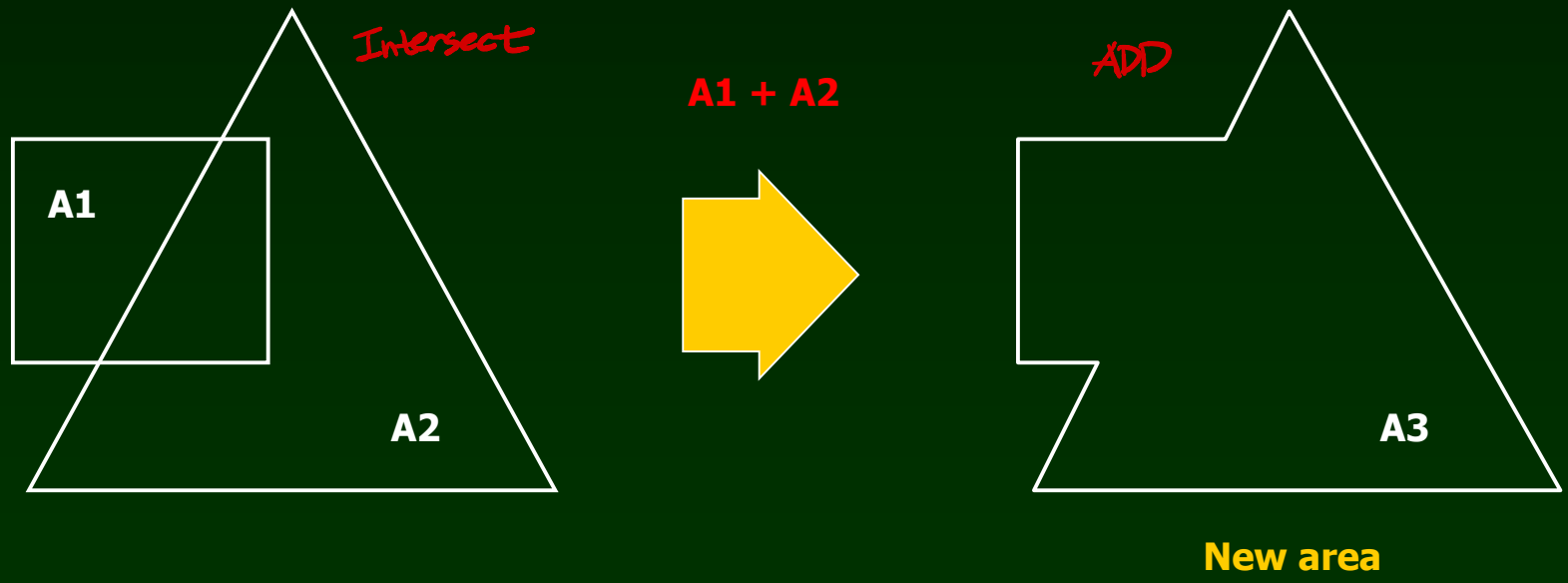


LINV (Line Intersect Volume)



Booleans - Add

- 수학적으로 보면 **Union, Joining** 혹은 **Summation**에 해당된다



Booleans - Add

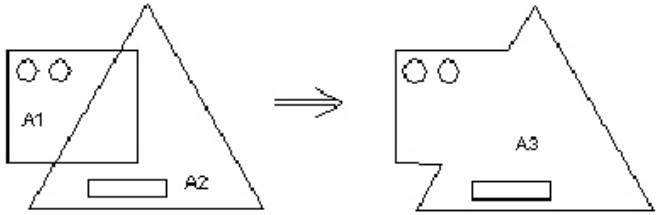
AADD, *NA1*, *NA2*, *NA3*, *NA4*, *NA5*, *NA6*, *NA7*, *NA8*, *NA9*

Adds separate areas to create a single area

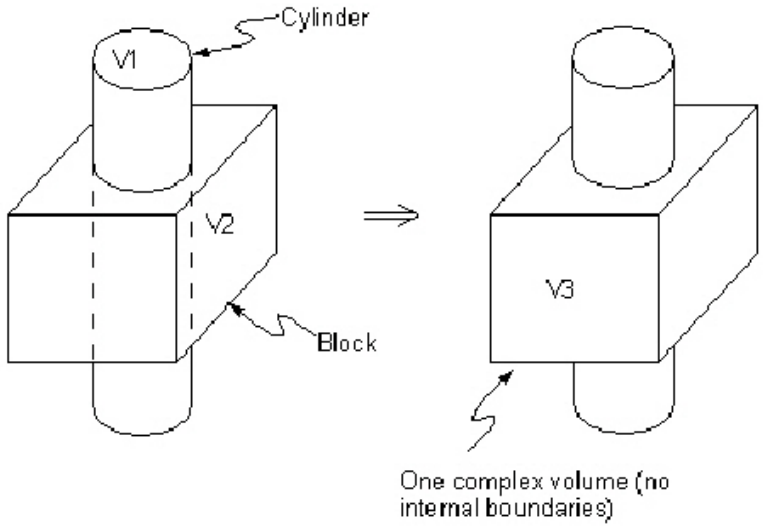
NA1*, *NA2*, *NA3*, *NA4*, *NA5*, *NA6*, *NA7*, *NA8*, *NA9

Numbers of areas to be added. If NA1 = ALL, add all selected areas and ignore NA2 to NA9. If NA1 = P, graphical picking is enabled and all remaining arguments are ignored (valid only in the GUI). A component name may also be substituted for NA1

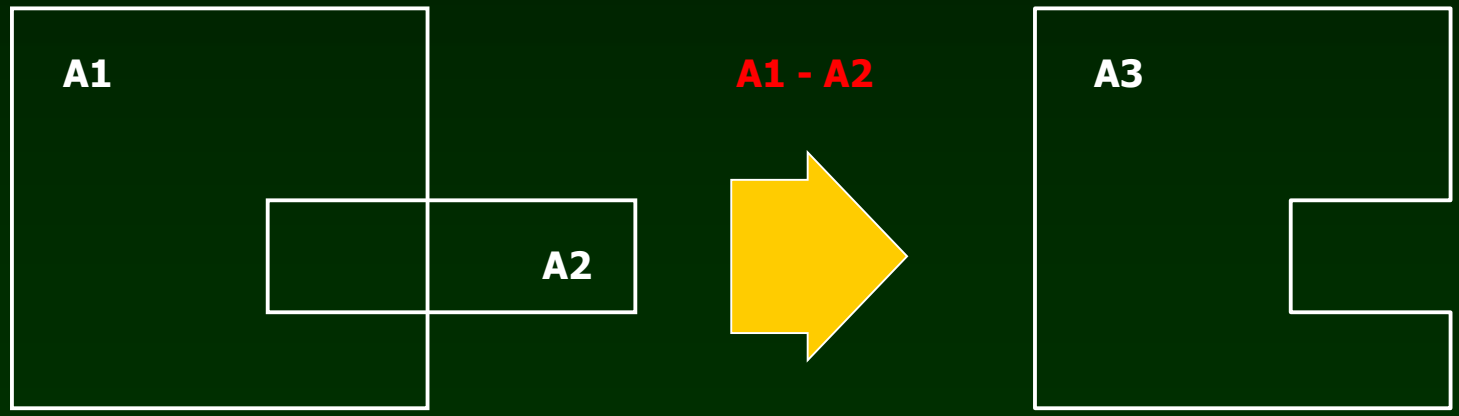
AADD (Add Areas)



VADD (Add Volums)



Booleans - Subtract



New area

Booleans - Subtract

ASBA, *NA1*, *NA2*, *SEPO*, *KEEP1*, *KEEP2*

Subtracts areas from areas.

A1에서 A2를 뺌.

NA1
Area (or areas, if picking is used) to be subtracted from. If ALL, use all selected areas. Areas specified in this argument are not available for use in the NA2 argument.

NA2
Area (or areas, if picking is used) to subtract. If ALL, use all selected areas (except those included in the NA1 argument).

SEPO
Behavior if the intersection of the NA1 areas and the NA2 areas is a line or lines:

(blank)	--	The resulting areas will share line(s) where they touch.
SEPO	--	The resulting areas will have separate, but coincident line(s) where they touch.

KEEP1
Specifies whether NA1 areas are to be deleted:

(blank)	--	Use the setting of KEEP on the BOPTN command.
DELETE	--	Delete NA1 areas after ASBA operation (override BOPTN command settings).
KEEP	--	Keep NA1 areas after ASBA operation (override BOPTN command settings).

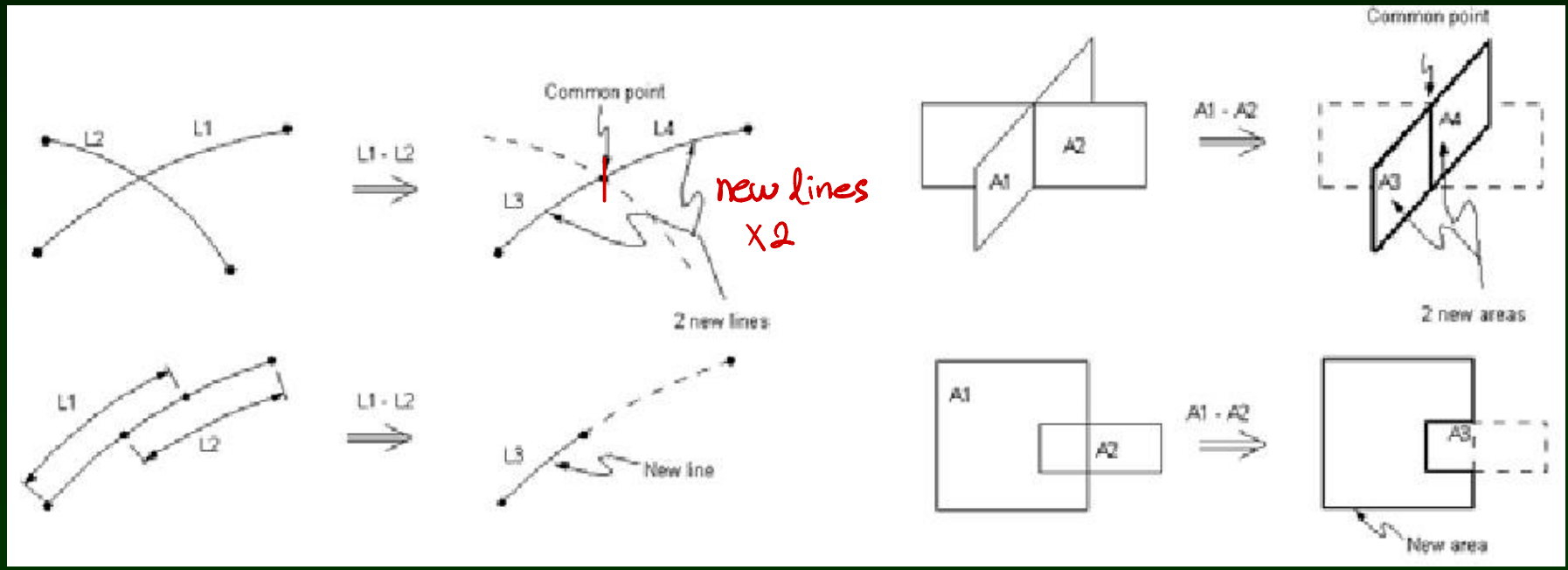
KEEP2
Specifies whether NA2 areas are to be deleted:

(blank)	--	Use the setting of KEEP on the BOPTN command.
DELET	--	Delete NA2 areas after ASBA operation (override BOPTN command settings).
KEEP	--	Keep NA2 areas after ASBA operation (override BOPTN command settings).

Booleans - Subtract

LSBL (Line Subtract Line)

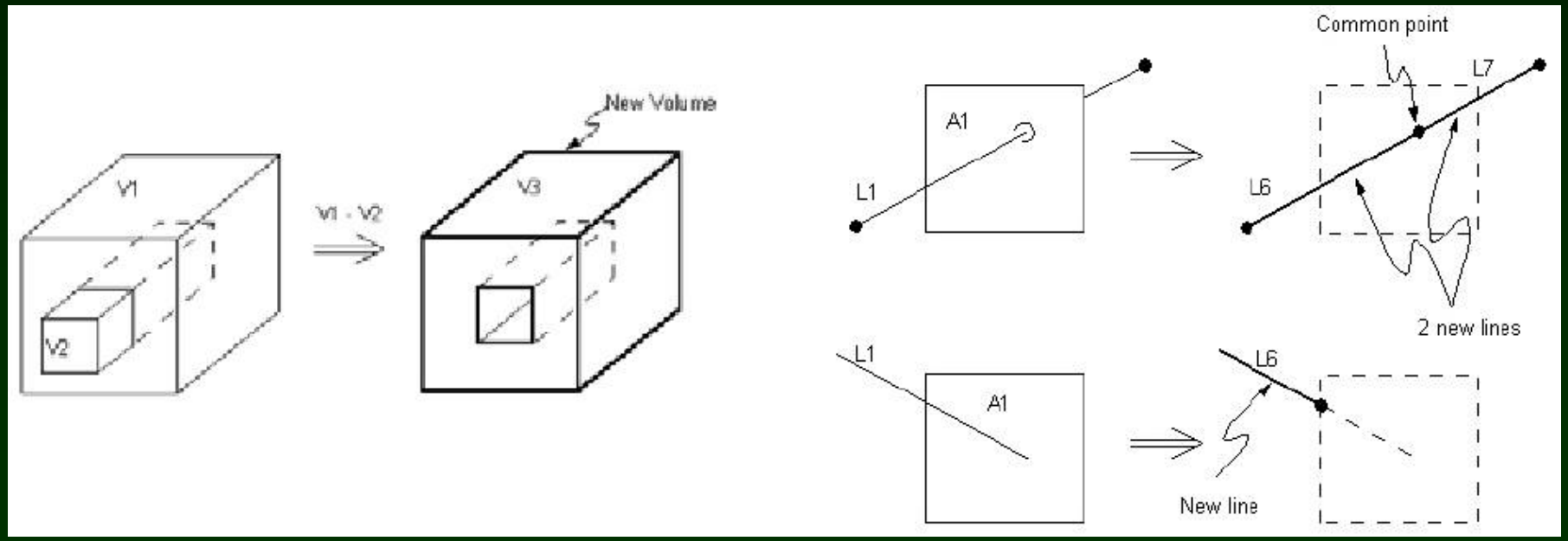
ASBA (Area Subtract Area)



Booleans - Subtract

VSBV (Volume Subtract Volume)

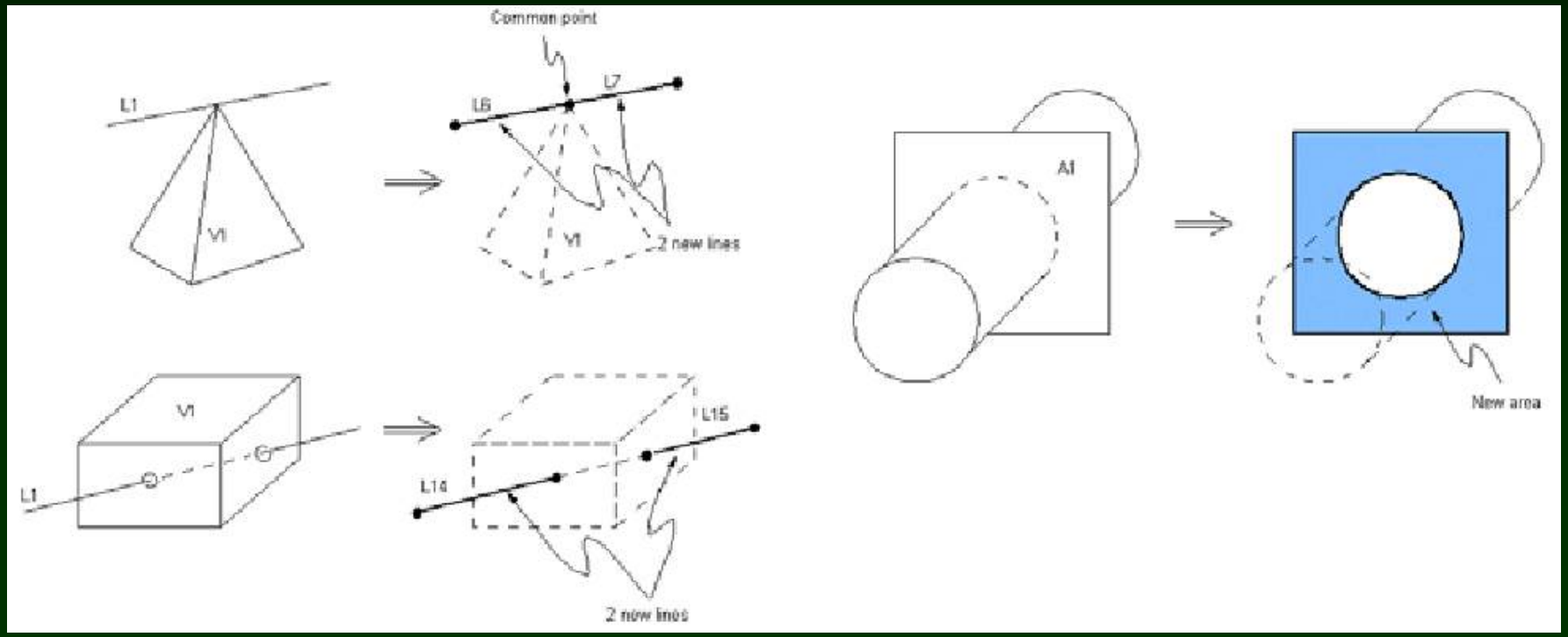
LSBA (Line Subtract Area)



Booleans - Subtract

LSBV (Line Subtract Volume)

ASBV (Area Subtract Volume)

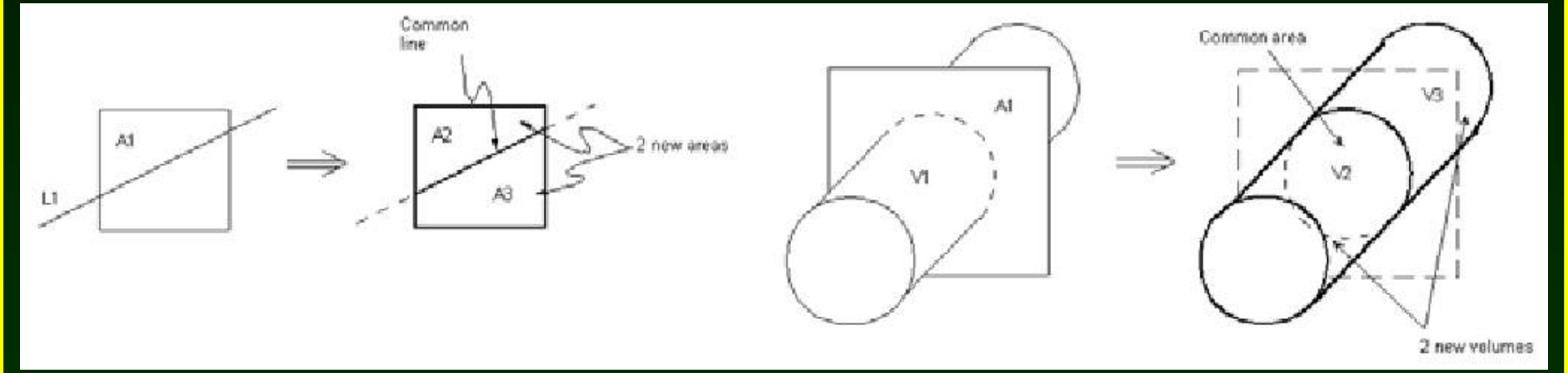


Booleans - Subtract

VSBA, 1, 1 : Volume '1' – Area '1'

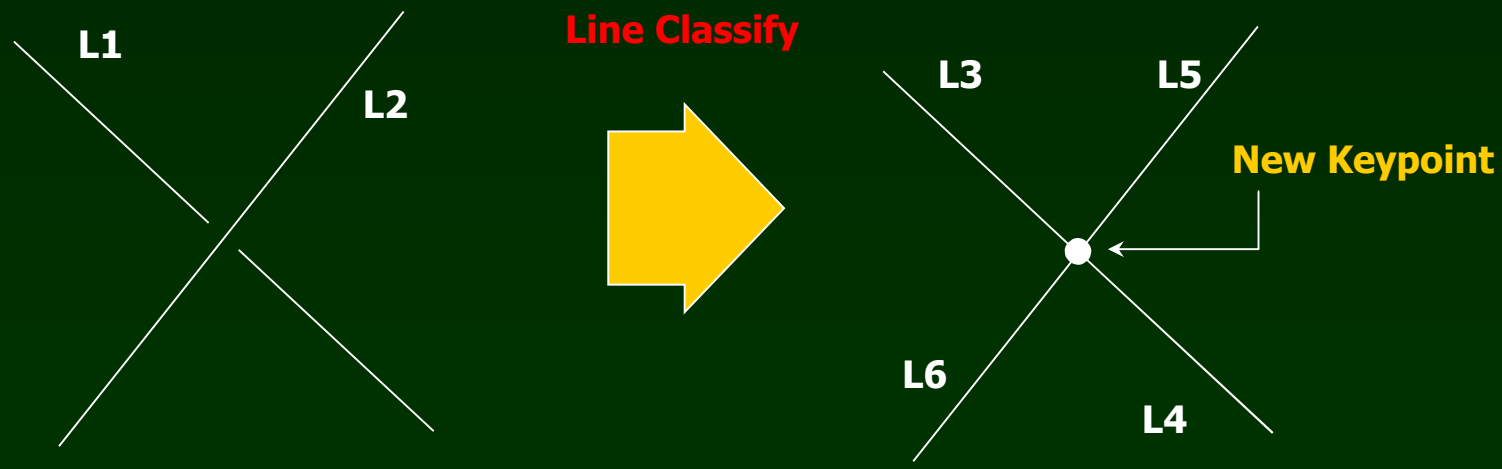
ASBL (Area Subtract Line)

VSBA (Volume Subtract Area)



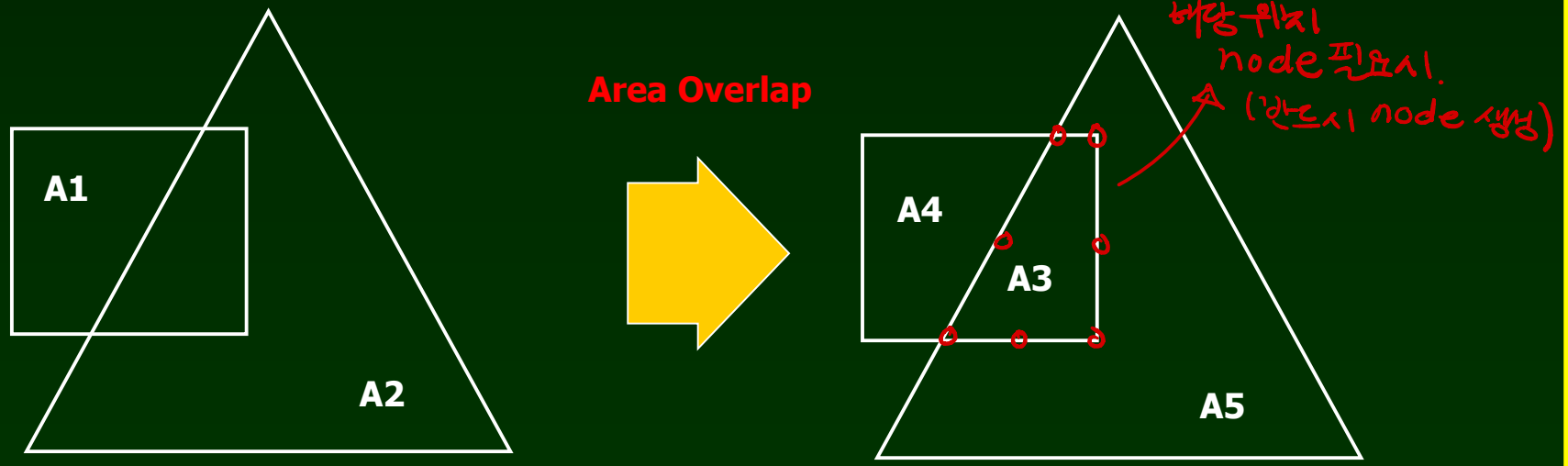
Line Classification

- Classification is similar to subtraction, except that both the original entities are replaced by new entities
- At present, only line-line classification is possible in the ANSYS program.



Booleans - Overlap

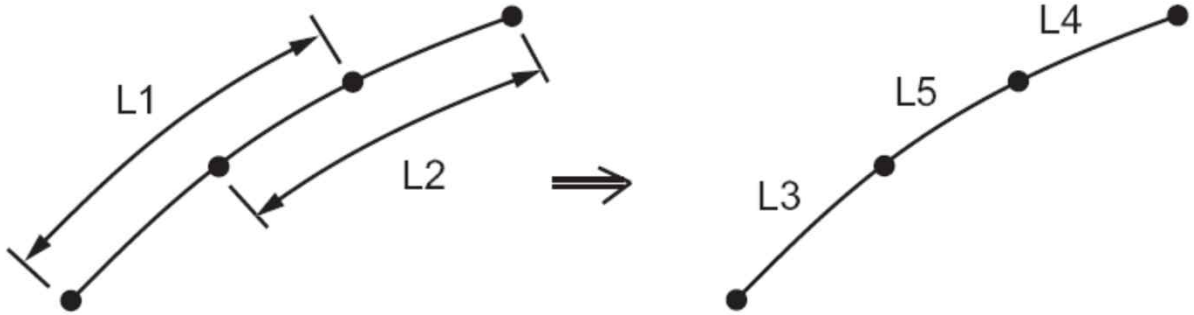
- Joining two or more entities to create three or more new entities that encompass all parts of the originals.
- The end result is similar to an "add" operation, except that boundaries will be created around the overlap zone.
- Thus, the overlap operation produces a number of relatively uncomplicated regions, as compared to the single relatively complicated region created by the add operation.



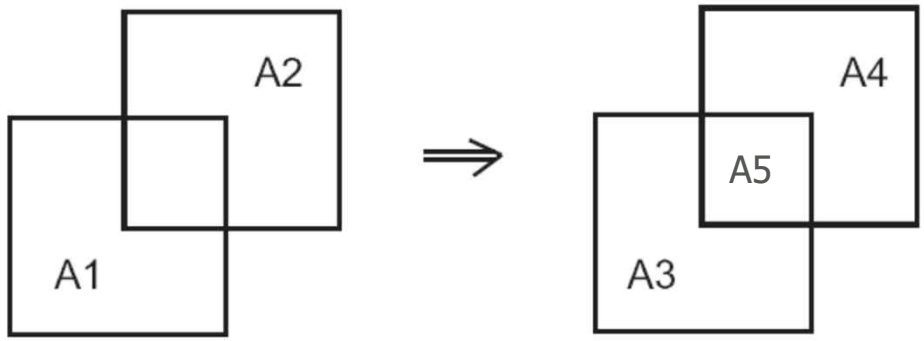
Booleans - Overlap

△ OVLAP
(L, A, ~~V~~)

LOVLAP (Line Overlap Line)

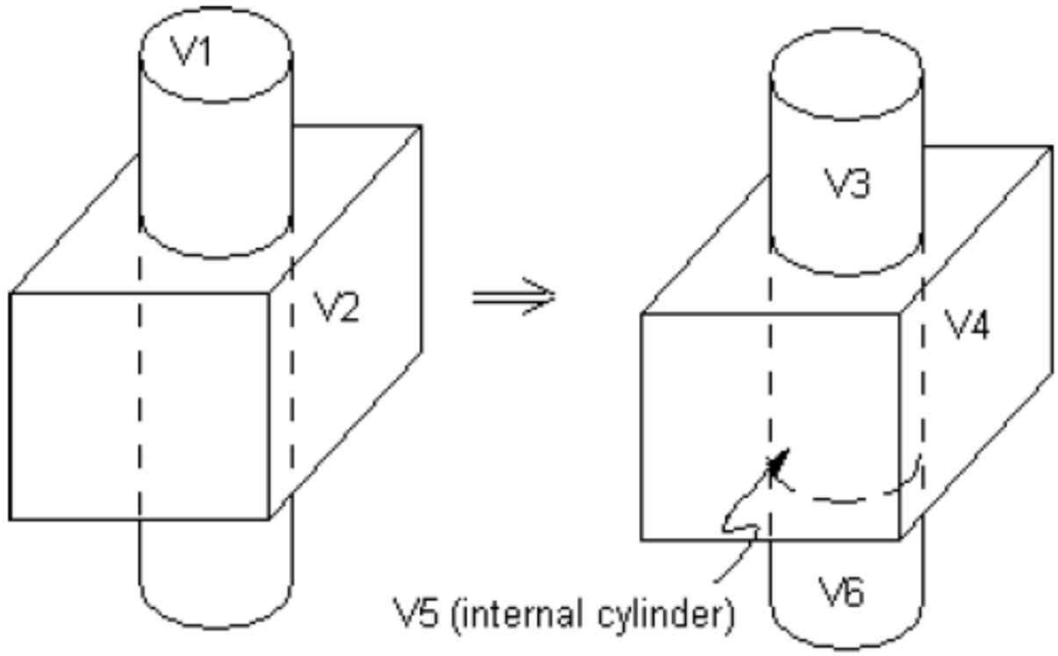


AOVLAP (Area Overlap Area)



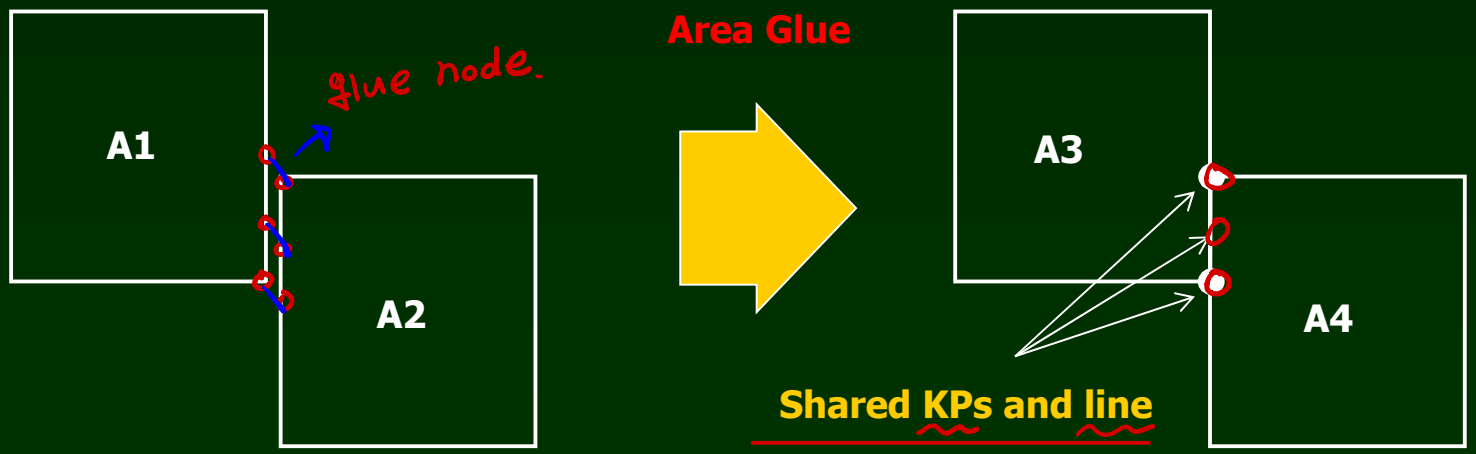
Booleans - Overlap

VOVLAP (Volume Overlap Volume)



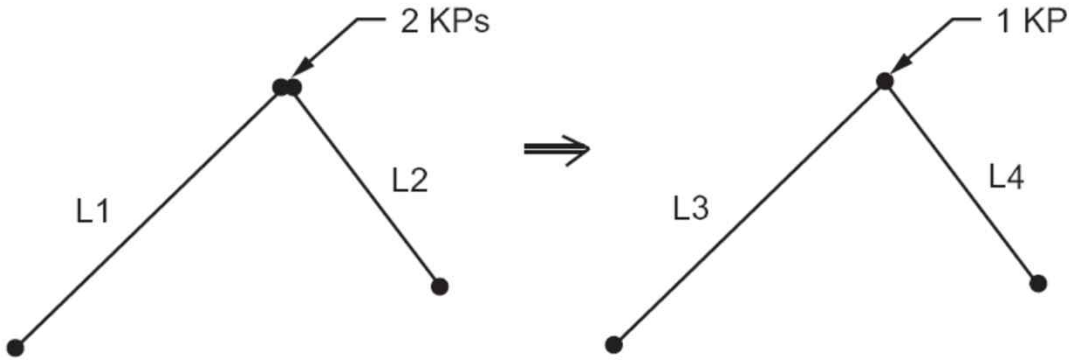
Booleans - Glue

- Glue is similar to overlap, except that it applies only to cases in which the intersection between entities occurs at a boundary, and is one dimension lower than the original entities.
- The entities maintain their individuality (they are not "added"), but they become connected at their intersection (they "talk" to each other)

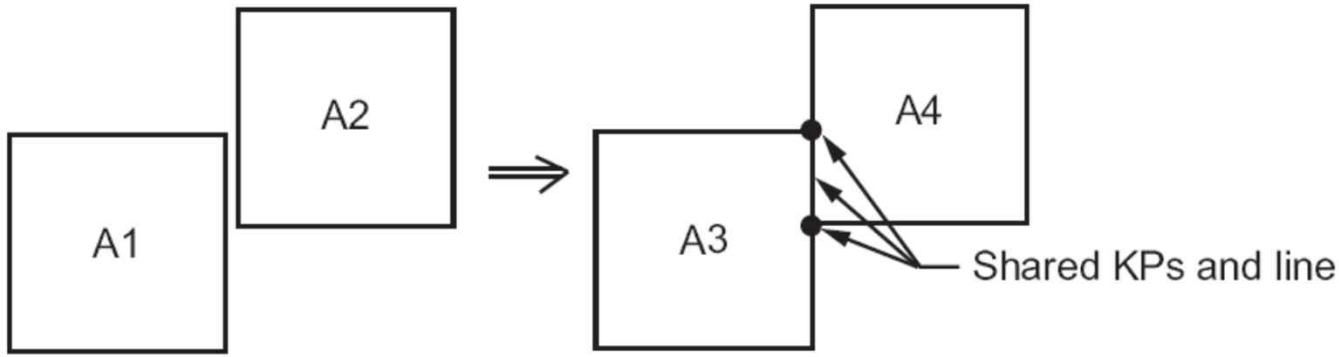


Booleans - Glue

LGULE (Line Glue Line)

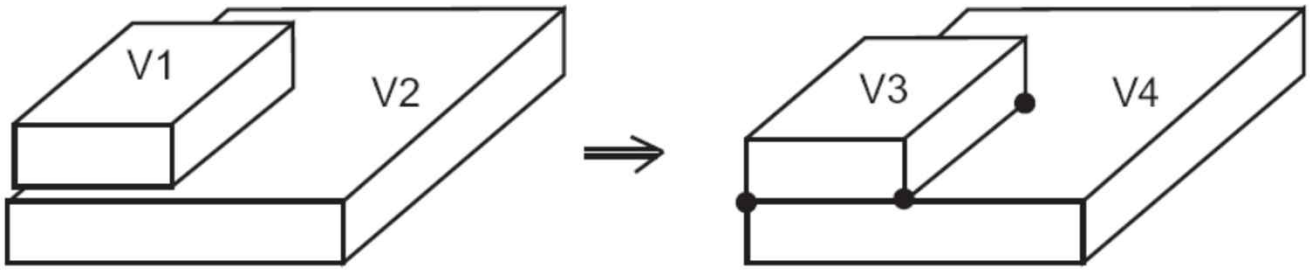


AGULE (Area Glue Area)



Booleans - Glue

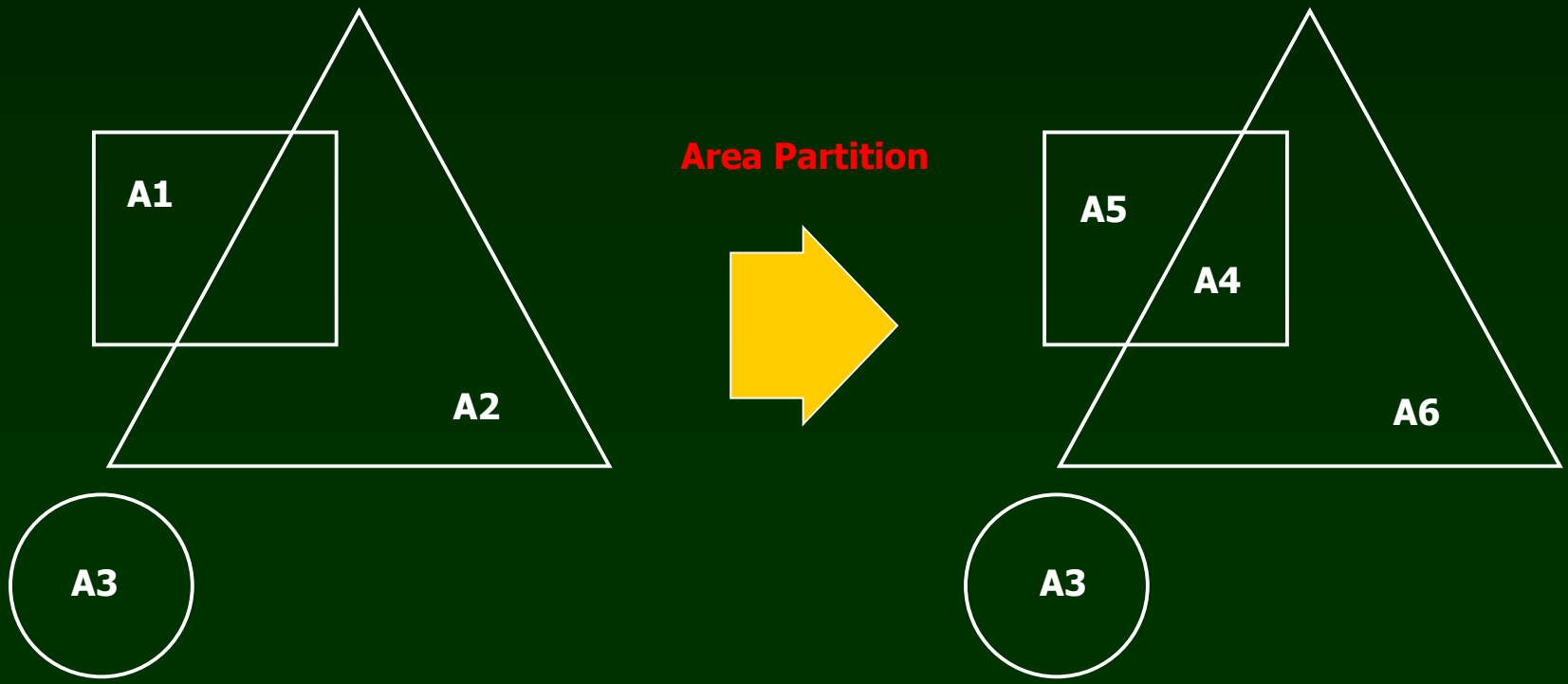
VGULE (Volume Glue Volume)



V3 and V4 share 4 keypoints,
4 lines, and an area

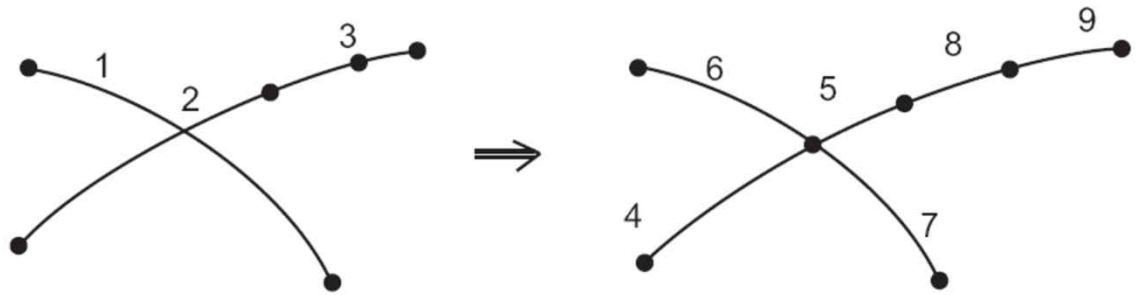
Booleans - Partition

- The end result is similar to an "overlap" operation if the overlap is of the same dimensionality as the original entities.
- However, unlike the overlap operations, non-overlapping input entities will not be deleted.

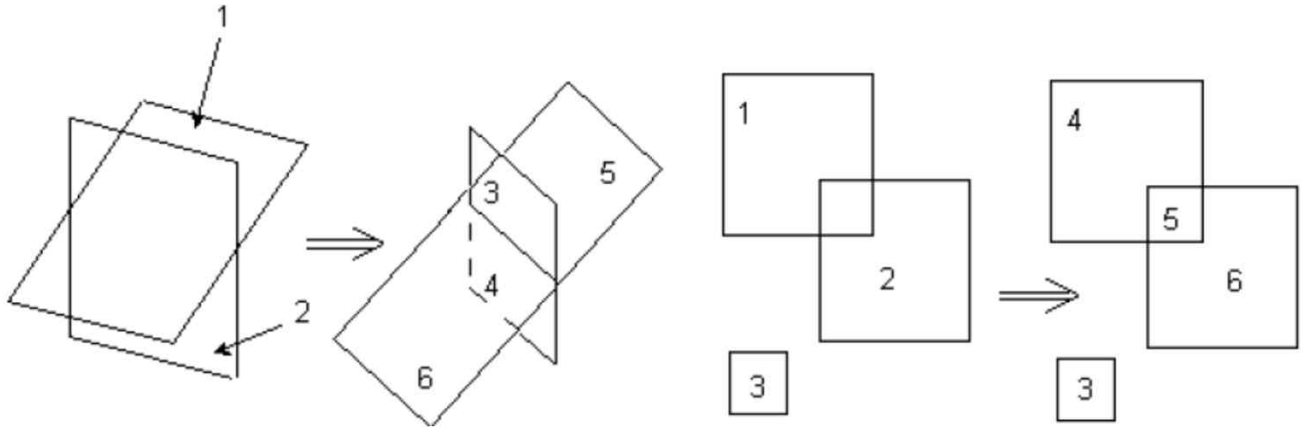


Booleans - Partition

LPTN (Line Partition)

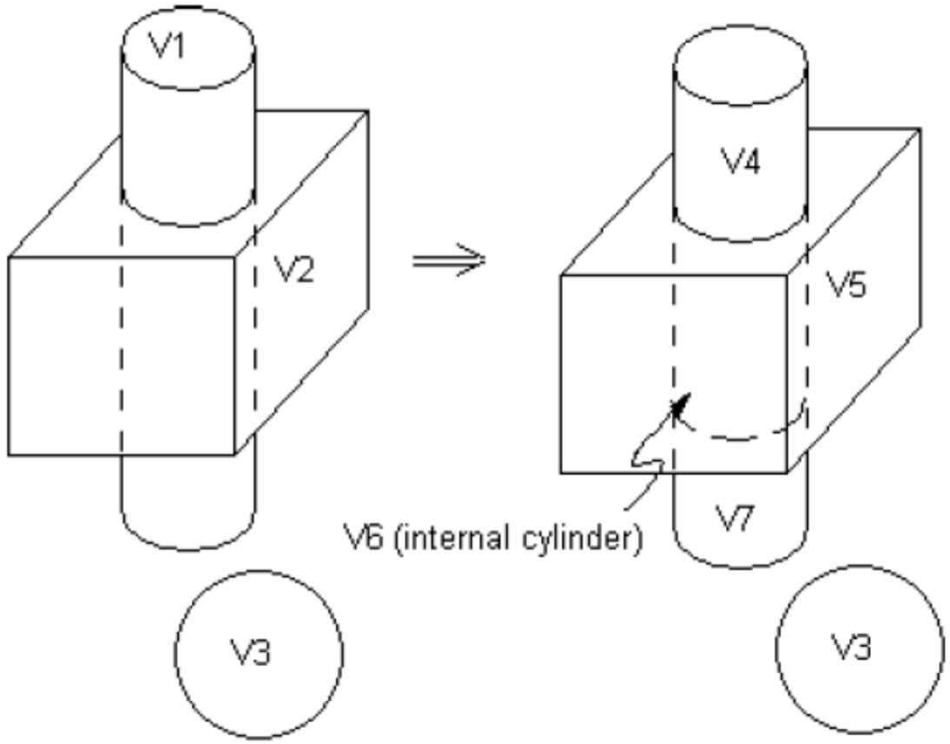


APTN (Area Partition)

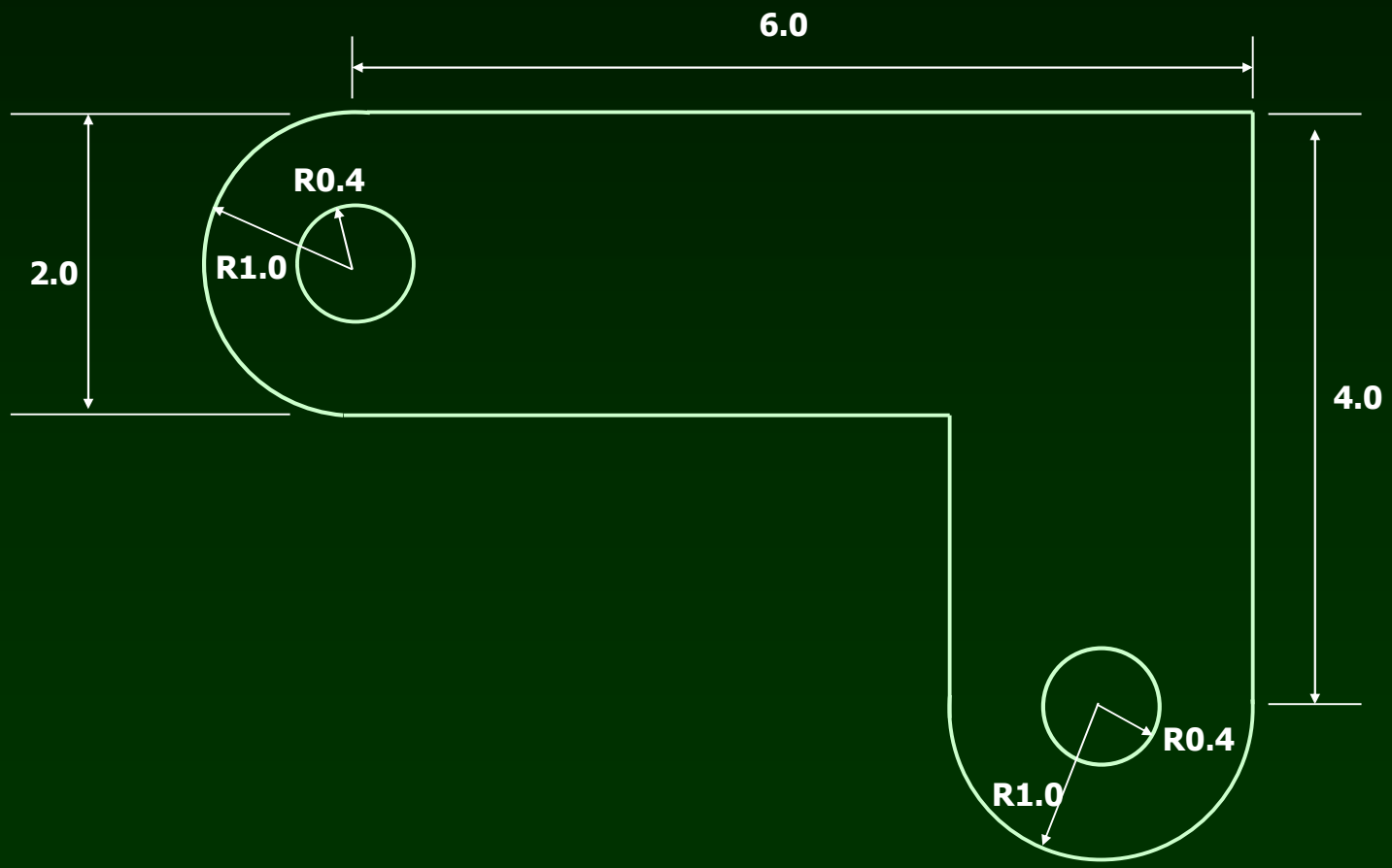


Booleans - Partition

VPTN (Volume Partition)



Exercise A



Meshing the Solid Model

〈저장〉
SAVE → file.db

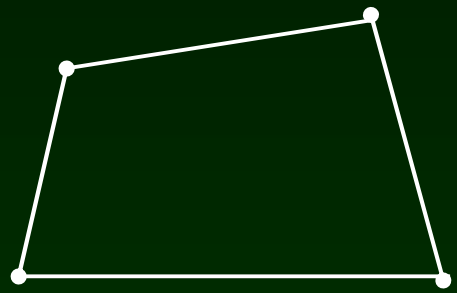
〈불러오기〉
Resume, filename, extension
RESUME, file, db

- **Meshing**에 필요한 정보
  - **Element Type**
  - **Geometric Property** ; Only 2D, 1D (3D는 필요x)
  - **Material Property**
  - **Element Size**

Element Type

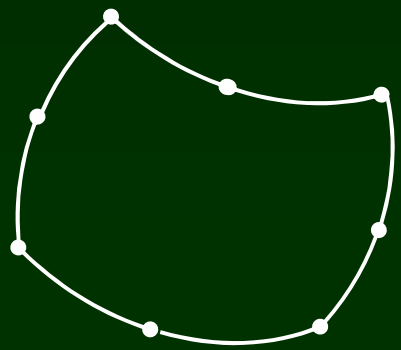
- **Element type**이 결정되면, 각 **Element**에 포함되는 **Node**수와 자유도가 결정된다.

PLANE182



2-D Structural Solid
4 Nodes
UX, UY degrees of freedom

PLANE183

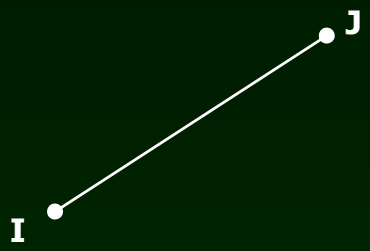


2-D Structural Solid
8 Nodes
UX, UY degrees of freedom

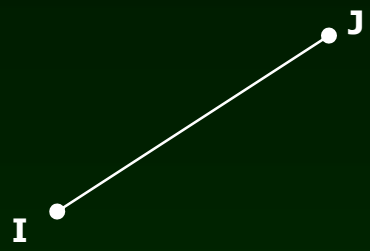
Element Type

BEAM
CIRCUit
COMBINation
CONTACt
FLUID
HF (High Frequency)
HYPERelastic
INFINite
INTERface
LINK
MASS
MESH
Multi-Point Constraint
PIPE
PLANE
PRETS (Pretension)
SHELL
SOLID
SOURCe
SURFace
TARGEt
TRANSducer
MATRIX
VISCOelastic (or viscoplastic)

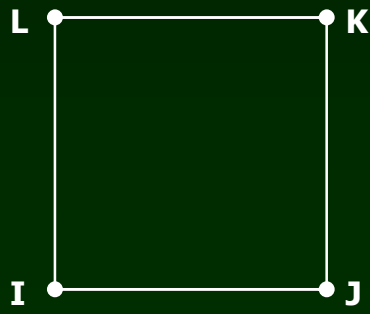
Element Type 정확한 ELEMENT TYPE 선택!



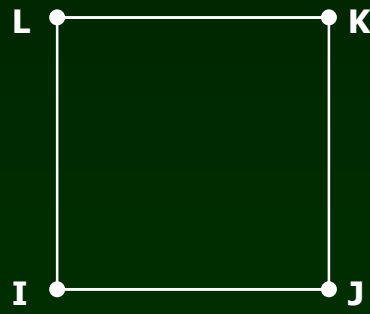
3-D Spar (Pin Joints)
UX, UY, UZ



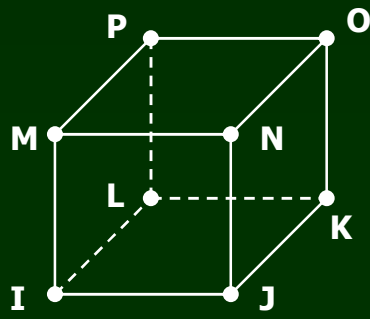
3-D Beam
UX, UY, UZ,
ROTX, ROTY, ROTZ



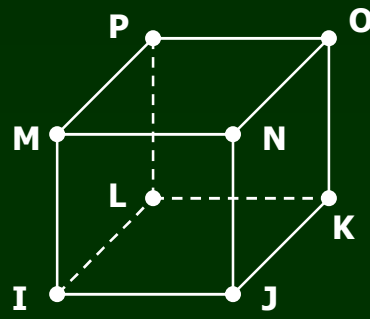
2-D or Axisymmetric Solid
UX, UY



3-D Quadrilateral Shell
UX, UY, UZ
ROTX, ROTY, ROTZ



3-D Structural Solid
UX, UY, UZ



3-D Thermal Solid
TEMP

Element Type

- **Element type**을 결정하는 **command**

ET, *ITYPE*, *Ename*, *KOP1*, *KOP2*, *KOP3*, *KOP4*, *KOP5*, *KOP6*, *INOPR*

Type reference number Set to 1 for now	Element name, e.g., LINK180, PLANE182

For example, ET, 1, PLANE182 **Specifies the element type to be PLANE182**

Element Type

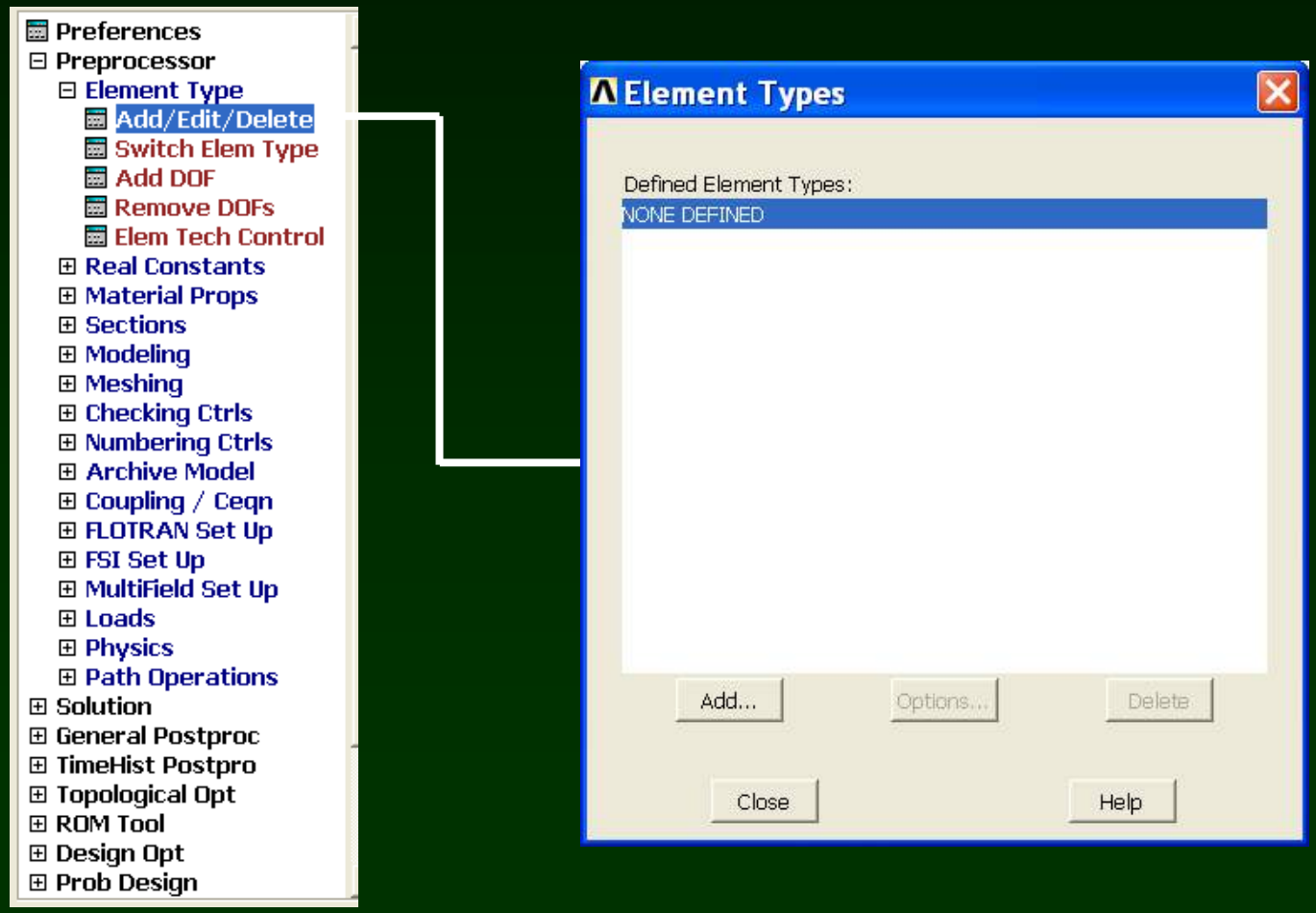
- Many element types have additional options, known as KEYOPTs, and are referred to as KEYOPT(1), KEYOPT(2), etc. e.g.:
  - KEYOPT(3) for BEAM188 allows you to choose shape functions along the length : 0 for linear(default), 2 for quadratic, 3 for cubic shape function on each element
  - KEYOPT(3) for PLANE182 allows you to set element behavior : 0 for plane stress, 1 for axisymmetric, 2 for plane strain, 3 for plane stress with thickness input
  - KEYOPT(1) for SHELL181 allows you to choose membrane option which takes translational degrees of freedom only

ET, ITYPE, Ename, KOP1, KOP2, KOP3, KOP4, KOP5, KOP6, INOPR

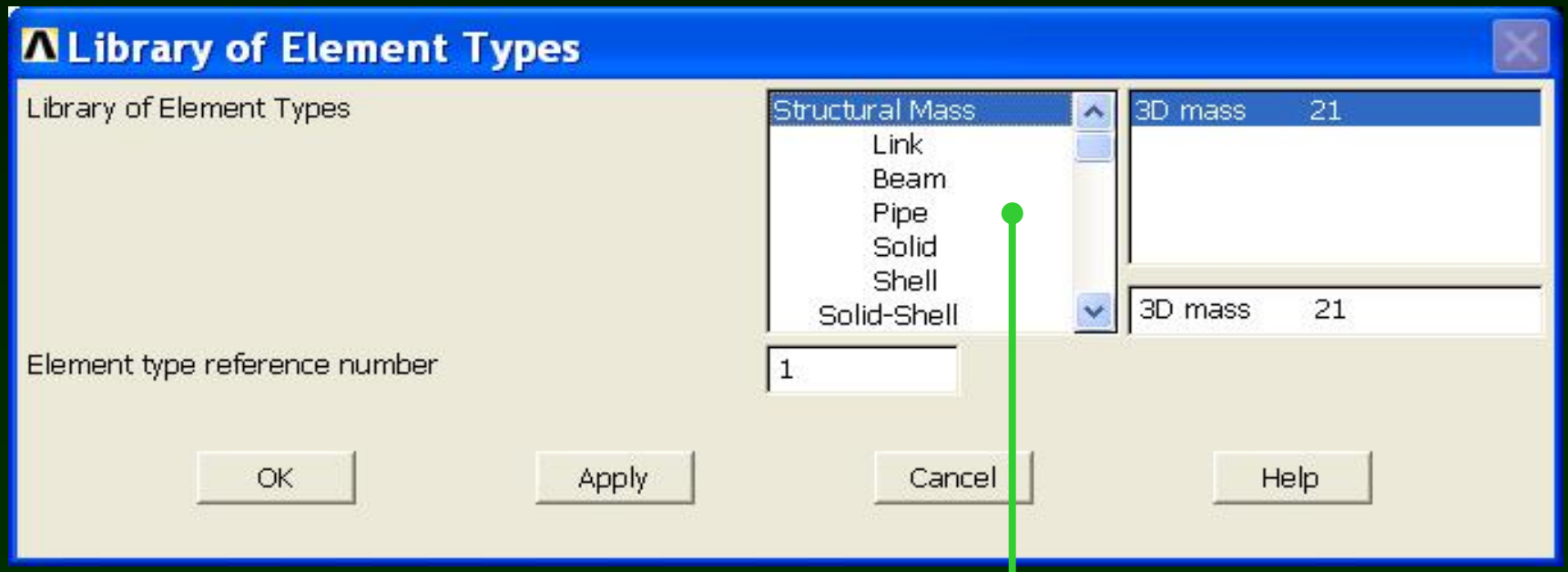
- KEYOPT values (1 through 6) for this element, as described in the ***ANSYS Elements Reference***.

Element Type

Main Menu> Preprocessor> Element Type> Add/Edit/Delete



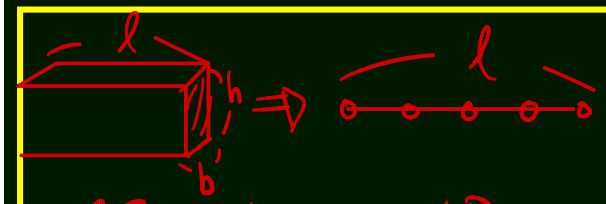
Element Type



The ANSYS element library contains more than 150 different element types

Each element type has a unique number and a prefix that identifies the element category

ET,1,BEAM188
ET,2,SHELL181

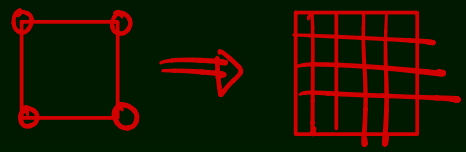


3D to 1D

(b, h 즉 A (단면적) 정보 삭제.

R (real constant) 이 SECT (Section) 에 비해 정보

Real Constants



Thickness 정보 필요.

- **Real constants are geometric properties specific to an element type**
  - ➢ **Cross-sectional properties of a beam element**
  - ➢ **Thickness of a shell element**
- **Not all element types require real constants, and different elements of the same type may have different real constant values.**
- **Real constant를 결정하는 command**

R : A, I, b, h

SECT : 단면, 2차 mom

단면형상.

R, *NSET*, *R1*, *R2*, *R3*, *R4*, *R5*, *R6*

Arbitrary reference number Set to 1 for now	**Real constants value, e.g., area, thickness, moment of inertia, etc.**

Real Constants

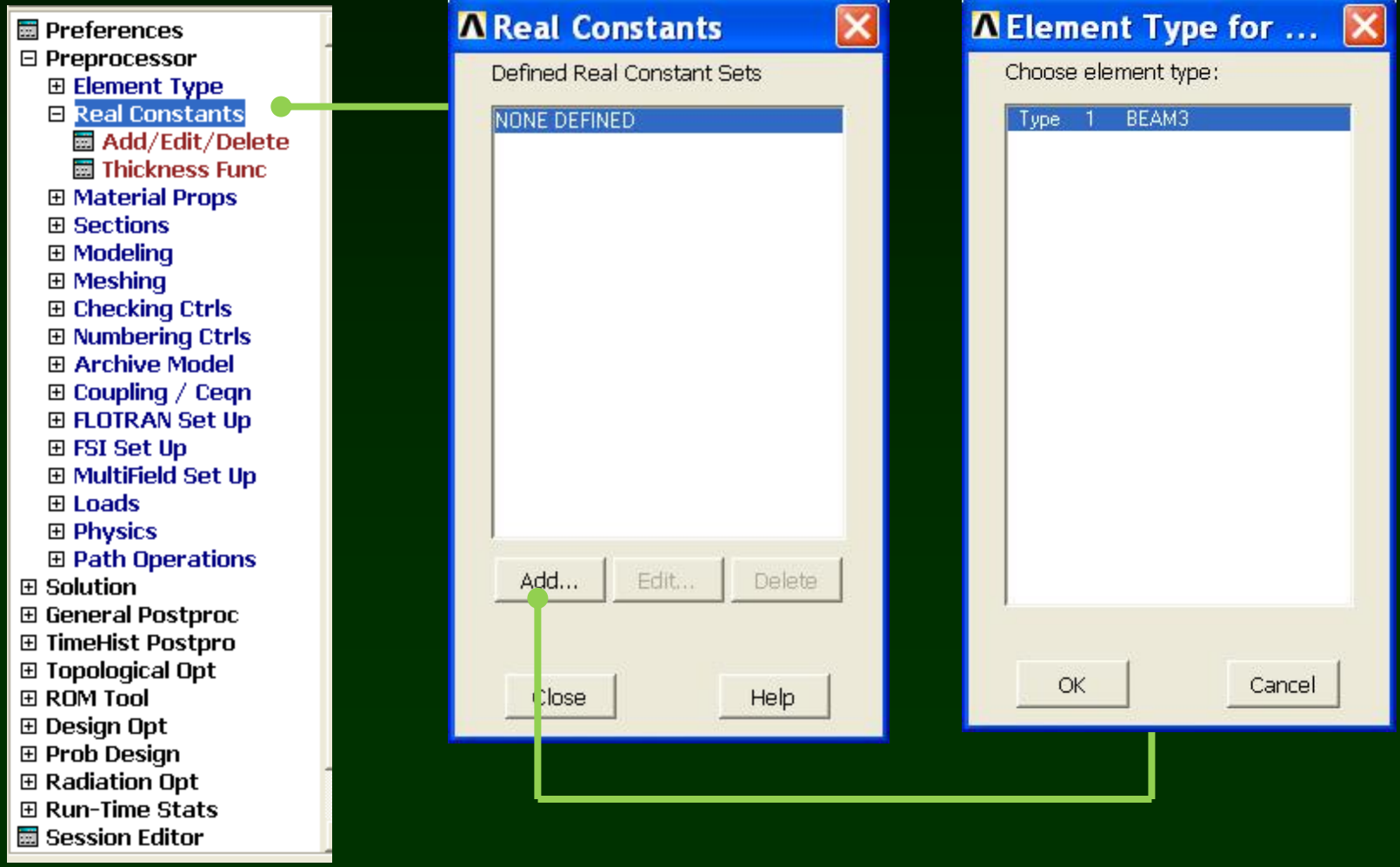
- For line and area elements that require geometry data (cross-sectional area, thickness, diameter, etc.) to be specified as real constants, you can verify the input graphically by using the following commands in the order shown:

  Utility Menu> PlotCtrls> Style> Size and Shape

  Utility Menu> Plot> Elements

- ANSYS displays the elements as solid elements, using a rectangular cross-section for link and shell elements and a circular cross-section for pipe elements. The cross-section proportions are determined from the real constant values.

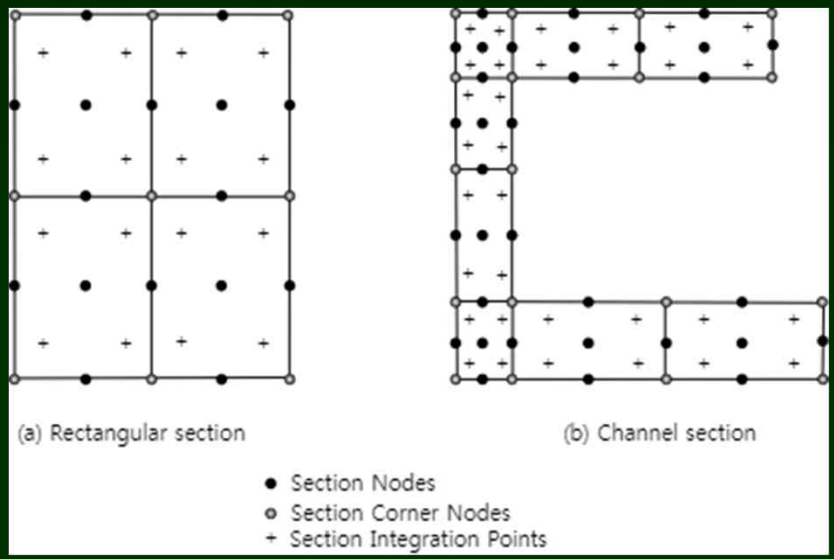
Real Constants

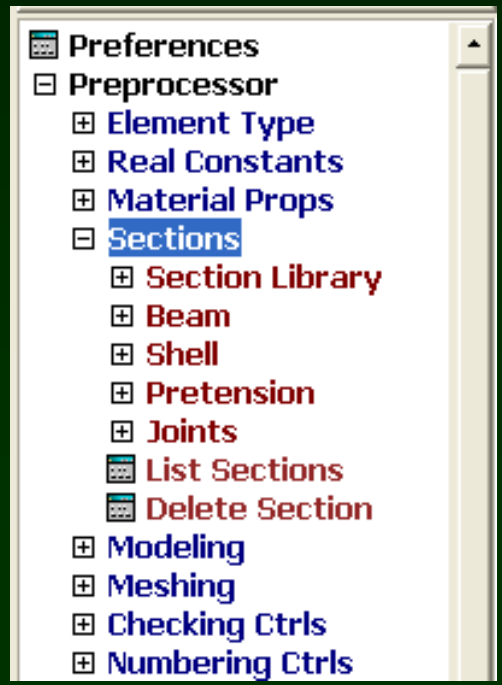


Sections

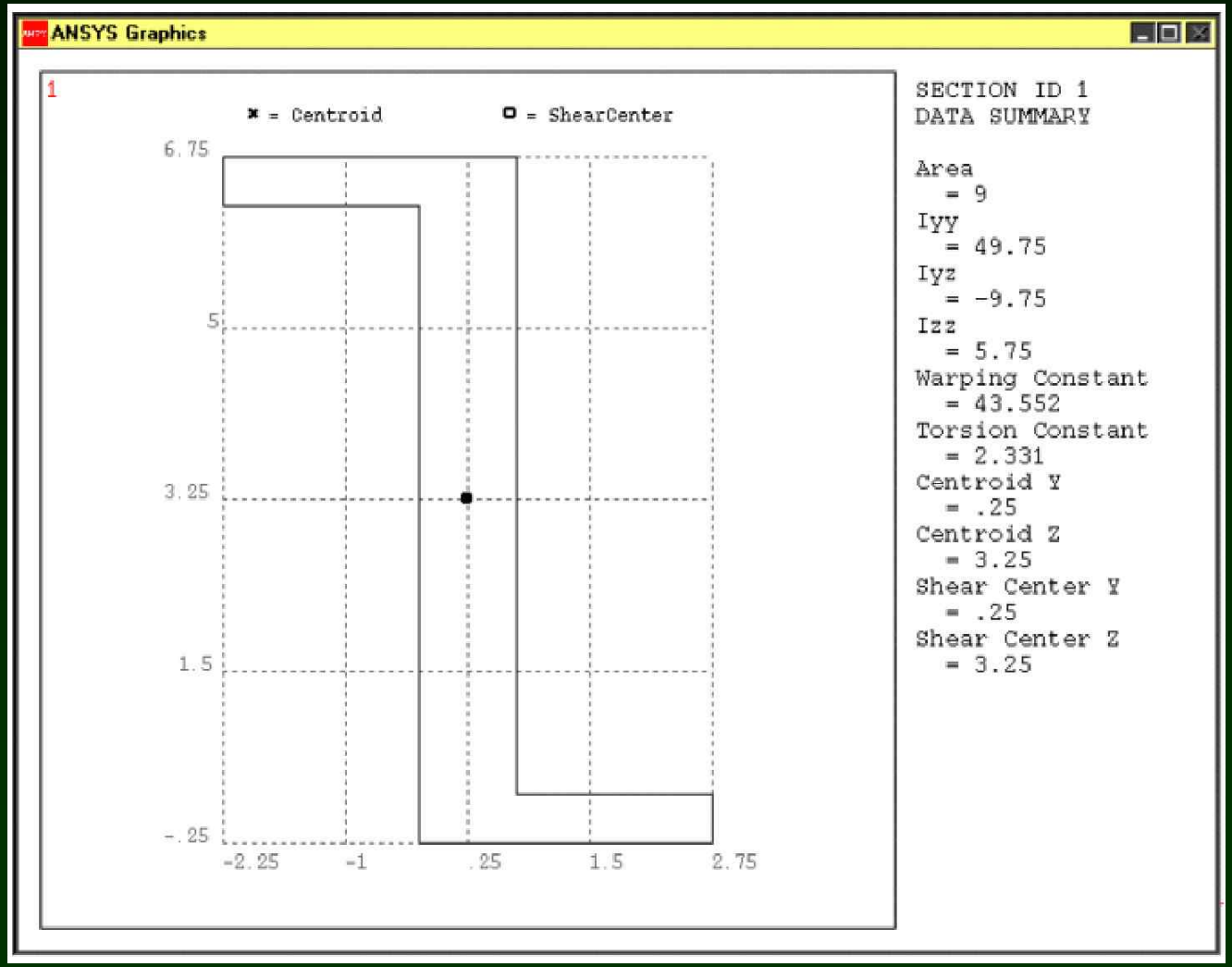
Section에서 단면형상 주면, I, A 등을 자동으로 계산.

- Building a model using BEAM44, BEAM188, or BEAM189, you can use the section commands (SECTYPE, SECDATA, etc.) or their GUI path equivalents to define and use cross sections in your models.
- A cross section defines the geometry of the beam in a plane perpendicular to the beam axial direction. ANSYS supplies a library of eleven commonly-used beam cross section shapes, and permits user-defined cross section shapes.
- When a cross section is defined, ANSYS builds a numeric model using a nine node cell for determining the properties (Iyy, Izz, etc.) of the section and for the solution to the Poisson's equation for torsional behaviour.

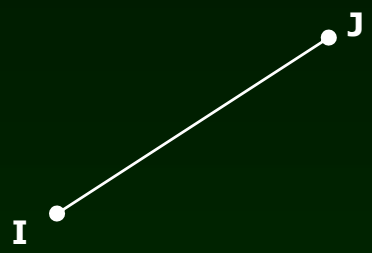




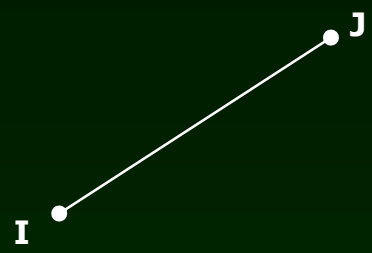
Sections


ANSYS Graphics
1
× = Centroid
□ = ShearCenter
6.75
5
3.25
1.5
-.25
-2.25
-1
.25
1.5
2.75
SECTION ID 1
DATA SUMMARY
Area
= 9
Iyy
= 49.75
Iyz
= -9.75
Izz
= 5.75
Warping Constant
= 43.552
Torsion Constant
= 2.331
Centroid Y
= .25
Centroid Z
= 3.25
Shear Center Y
= .25
Shear Center Z
= 3.25

Sections



3-D Spar (Pin Joints)
UX, UY, UZ



3-D Beam
UX, UY, UZ,
ROTX, ROTY, ROTZ

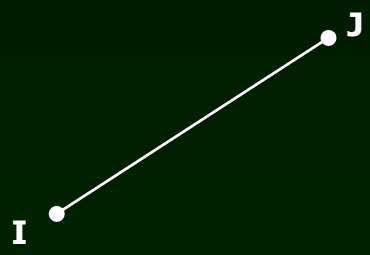
- **LINK180 : The element is defined by two nodes, the cross-sectional area (A) input via the SECTYPE and SECDATA commands.**
- **BEAM188 : The cross-section details are provided separately via the SECTYPE and SECDATA commands. (See Beam and Pipe Cross Sections in the Structural Analysis Guide for details.)**

- **Section type을 결정하는 command**

SECTYPE, SECID, Type, Subtype, Name, REFINEKEY

- **SECTYPE sets the section ID number, section type, and subtype for a section. The geometry data describing this section type is defined by a subsequent SECDATA command. The SLIST command lists the section properties, and the SECPLOT command displays the section to scale.**

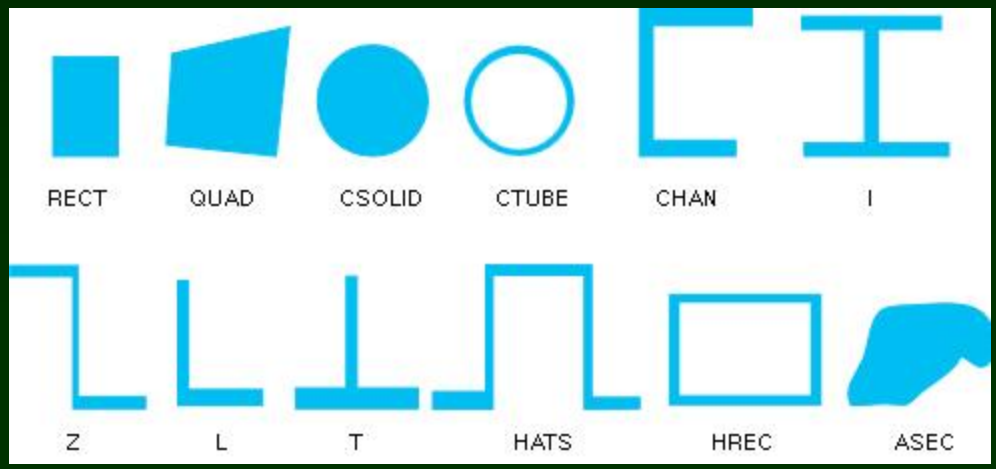
Sections



3-D Beam
UX, UY, UZ,
ROTX, ROTY, ROTZ

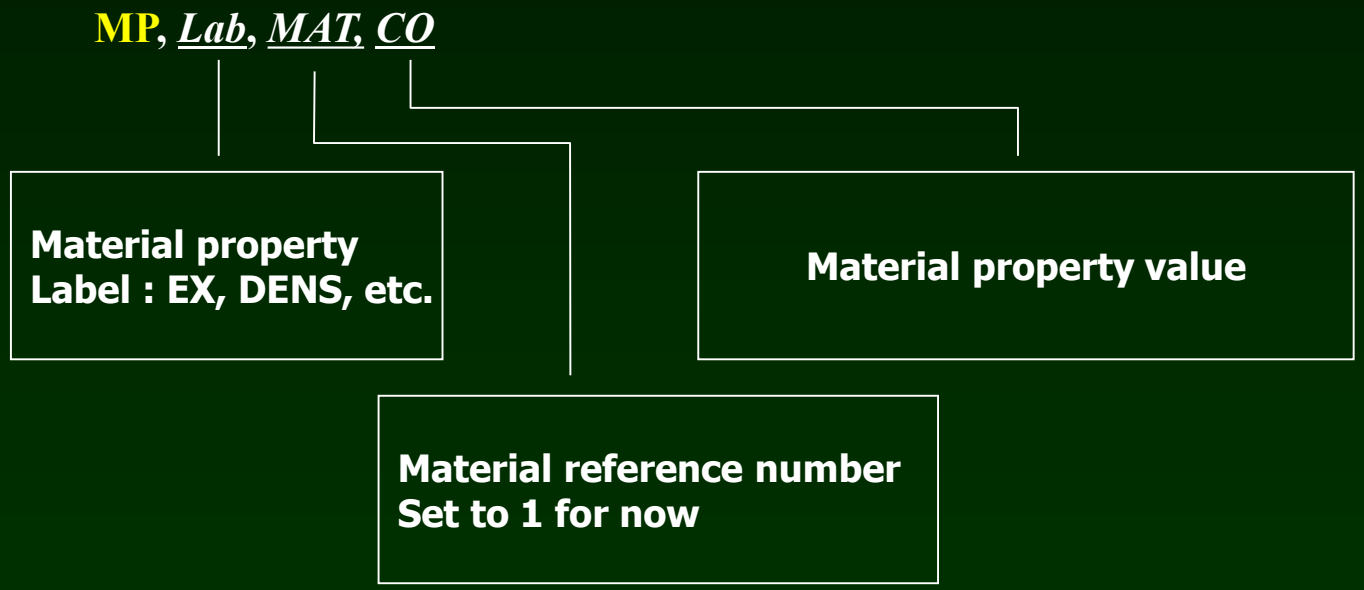
SECTYPE, SECID, Type, Subtype, Name, REFINEKEY

- When Type = BEAM, the possible beam sections that can be defined for Subtype are :



Material Property

- **Material property**를 결정하는 **command**



Material Property

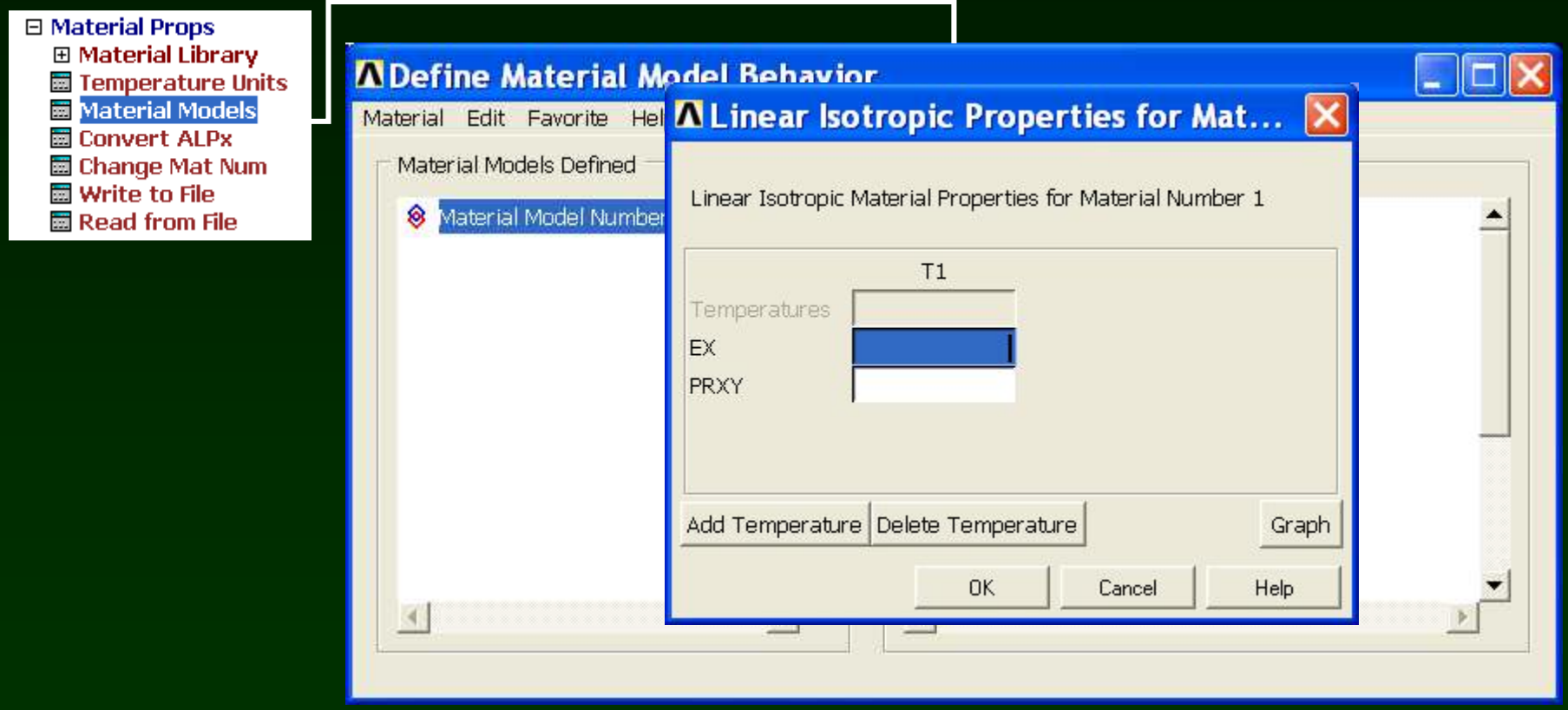
• 해석 종류에 따른 필요한 물성치 예

➢ **Structural**	**Young's modulus (EX)** **Poisson's ratio (NUXY)** **Density (DENS)** **Etc.**
➢ **Thermal**	**Thermal conductivity (KXX)** **Specific heat (C)** **Etc.**
➢ **Magnetic**	**Relative permeability (MURX)** **Permittivity (PERX)** **Etc.**

Isotropic, 가로세로 같다.

Isotropic.

Material Property




```
MP,EX,1,2E11      ! Young's modulus for material ref. no. 1 is 2E11
MP,NUXY,1,0.3     ! Poisson's ratio for material ref. no. 1 is 0.3
MP,DENS,1,7800    ! Density for material ref. no. 1 is 7800
```

Element Size

Ex3.1 Top-Down. : LSize

- **Element size**를 결정하는 **command**

Line 당 Element 하나씩 만들수 있다.

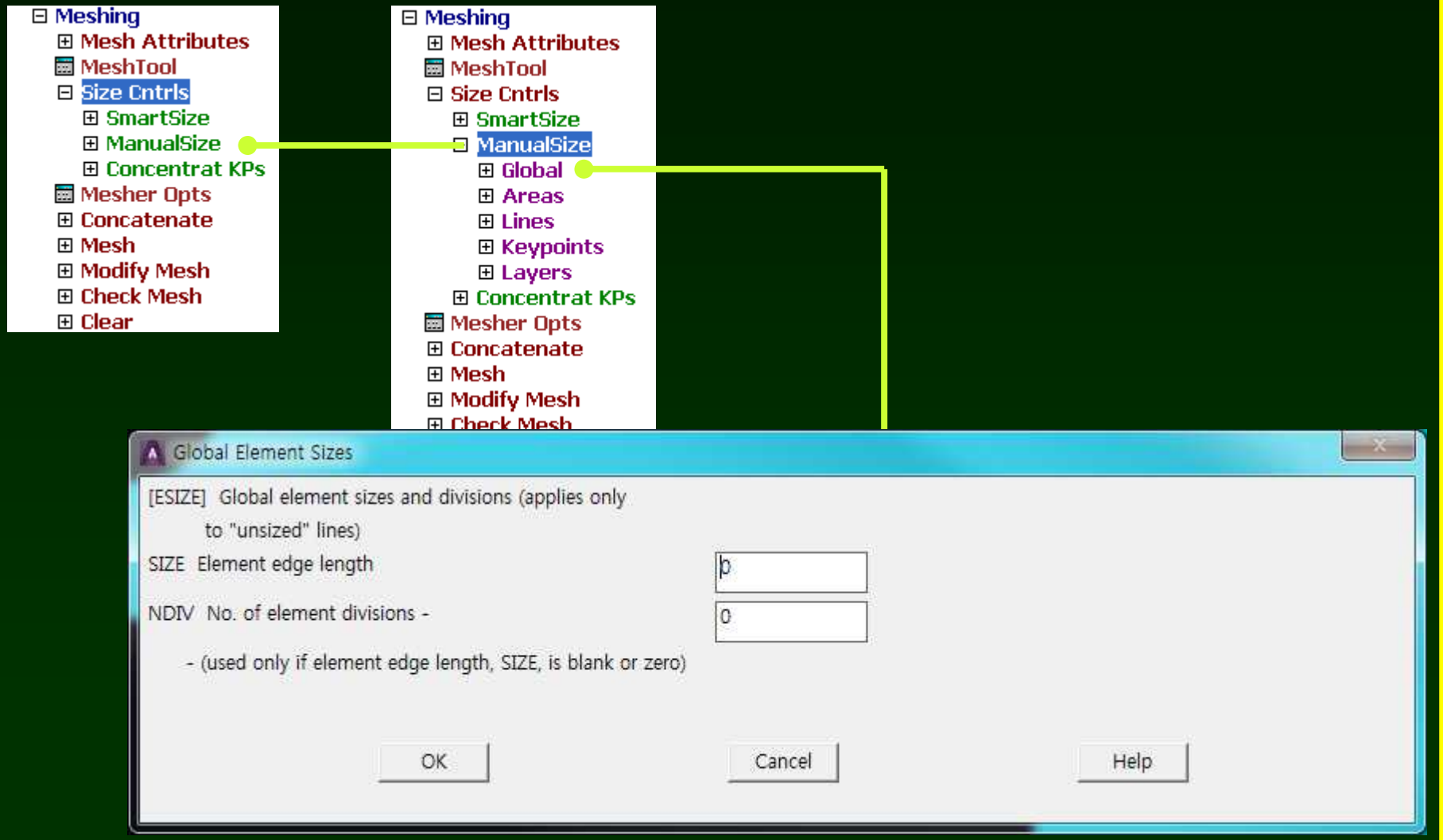
ESIZE, ***SIZE***, … NDIV

Element edge length.
Units should be consistent with the rest of the model data.

For example, ESIZE, 1

Specifies the element size to be 1.0 unit, and should not exceed 1.0 unit over all element edge length

Element Size



Save

- **Save command**

SAVE, *Fname*, *Ext*, …

File name prefix. Defaults to FILE (the jobname)	File name extension. Defaults to DB (for database) if *Fname* is blank

- **Resume command**

RESUME, *Fname*, *Ext*, …

Meshing

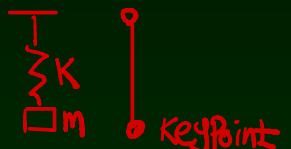
Solid Model - Geometry

- **Mesh를 위한 command**

➢ **To generate point elements (such as MASS21) at keypoints :**

✓ **Command(s): KMESH**

GUI : Main Menu> Preprocessor> Meshing> Mesh> Keypoints



➢ **To generate line elements (such as BEAM188) on lines :**

✓ **Command(s): LMESH**

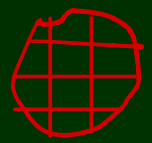
GUI : Main Menu> Preprocessor> Meshing> Mesh> Lines



➢ **To generate area elements (such as PLANE182) on areas :**

✓ **Command(s): AMESH**

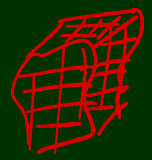
GUI : Main Menu> Preprocessor> Meshing> Mesh> Areas



➢ **To generate volume elements (such as SOLID185) in volumes:**

✓ **Command(s): VMESH**

GUI : Main Menu> Preprocessor> Meshing> Mesh> Volumes



Meshing

Aclear : 내부 NODES 삭제.
Adelete : Area 관계 삭제
□CLEAR VS □DELETE

- **Area Mesh를 위한 command**

AMESH, *NA1*, *NA2*, *NINC*

Area number of the area to be meshed or ALL or PICK	**Ending area number and number increment. Used for meshing multiple areas. *NINC* defaults to 1.**

For example, AMESH, 1, 5, 2

Specifies the area numbers to be meshed are 1, 3, and 5.

Area 와 Element
DETACH

ACLEAR, *NA1*, *NA2*, *NINC*

Deletes nodes and area elements associated with selected areas.

et, 1, Plane182
mp, ex, 1, 30e6
mp, dens, 1, 0.238
mp, nuxy, 1, 0.3

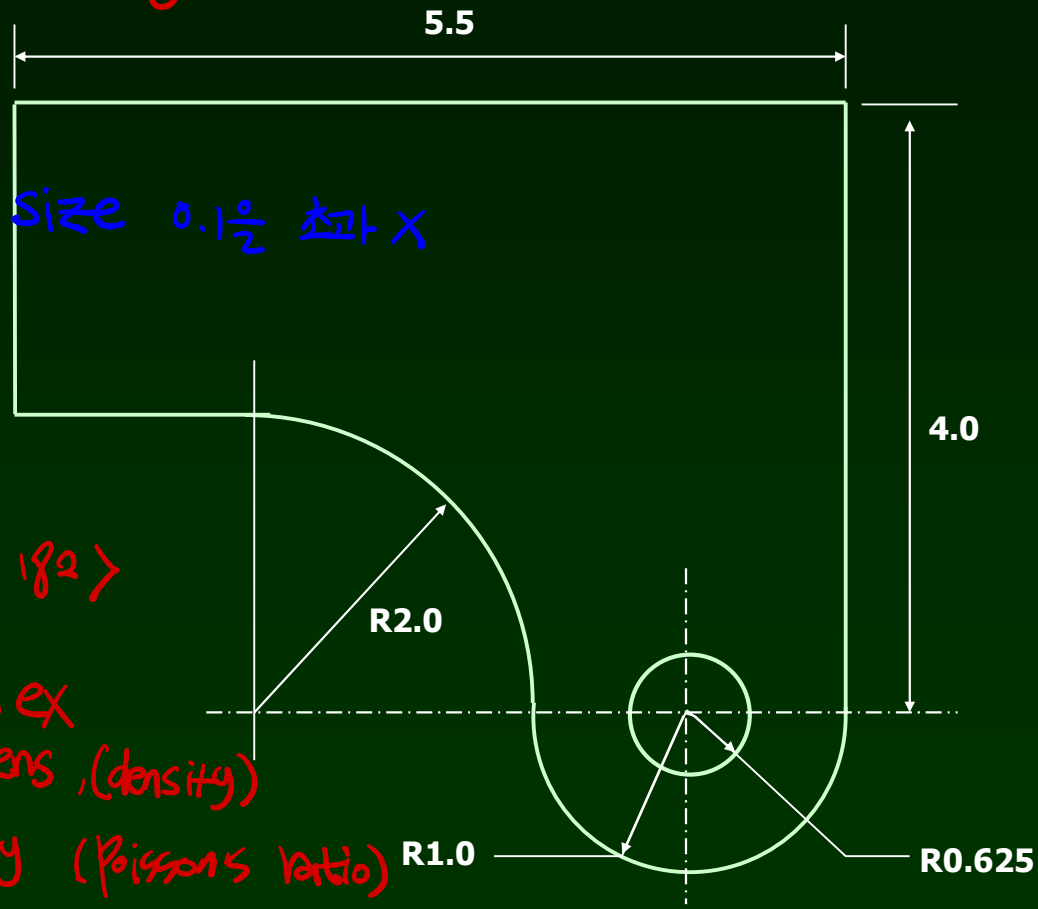
material property

nlist, elist.

Exercise B

esize, 0.1
amesh, 2

element size 0.1은 최소 X



<Plane 182>

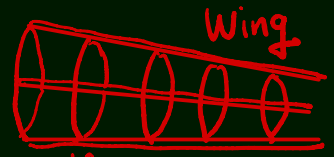
Material properties :

$E = 30 \times 10^6$ lb$_f$/in^2 , ex

$\rho = 0.238$ lb/in^3 , dens (density)

$\nu = 0.3$, nuxy (Poissons ratio)

Element Attributes



DEFINE) ET R MT SECTYPE

(ALL)

	TYPE
1	PLANE182
2	BEAM188
P	

	REAL
1	A1, I1, H1
2	A2, I2, H2
3	Thickness
Q	

	MAT
1	Steel
2	Aluminum
R	

	SECNUM
1	AXIS
2	BEAM
3	LINK
S	

Reference Number

2elements

Skin : Shell , ET, 1, Shell
Rib : Solid , ET, 2, Solid
SPAR : BEAM188 , ET, 3, BEAM188

Element 3개 中 / 재정 ⇒ Mesh.

ex) 2번 element 재정후 mesh.

`TYPE, 2`
`LMESH`

- Set the element type attribute pointer

  TYPE, *ITYPE* ⟶ Type reference number

  ⇒ ET로 재정했던 ref. num.

- Set the element real constant attribute pointer

  REAL, *NSET* ⟶ Real constant set number

- Set the element material attribute pointer

  MAT, *MAT* ⟶ Material reference number

- Set the element section attribute pointer

  SECNUM, *SECID* ⟶ Section ID number

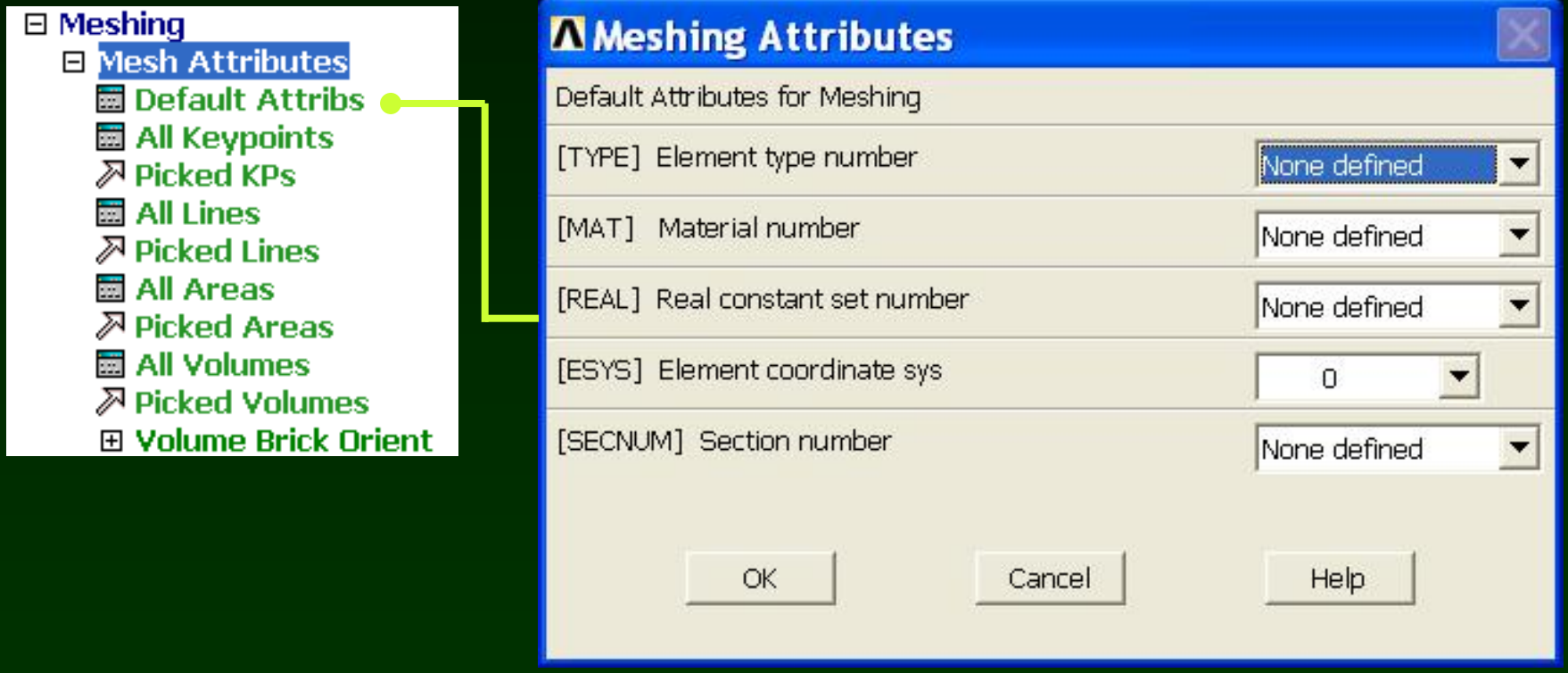
Type 3
MAT 2
Vmesh, 10 → Activated
Type 3, MAT2

Type 1
AMesh, 5 → Activated
Type 1, MAT2

Type 2
MAT 1
LMesh, 3

Show, vec : LineEdge

Element Attributes



Load Categories

- DOF Constraints → B.C. (fixed or free)
- Point Loads → Concentrated force
- Surface Loads → Pressure
- Body Loads
- Inertia Loads

DOF Constraints , B.C.

- Fixed displacements (zero or non-zero values) in a structural analysis.
  Most DOF constraints are used to:

  - Enforce symmetry conditions or so-called "built-in" boundaries.

  or

  - Prevent rigid body motions.

- Prescribed temperatures in a thermal analysis.

DOF Constraints

❑ **DOF Constraints**

- **D : Defines DOF constraints at nodes**
- **DA : Defines DOF constraints on areas**
- **DL : Defines DOF constraints on lines**
- **DK : Defines DOF constraints at keypoints**
- ***x*LIST : Lists DOF constraints**
- ***x*DELE : Deletes DOF constraints**

1번 node 'fully fixed'

D, 1, ALL, Ø

1~4번 nodes 'fully fixed'

NEND 사용

- Define DOF constraints at nodes

D, *Node*, *Lab*, *VALUE*, *VALUE2*, *NEND*, *NINC*, *Lab2*, *Lab3*, *Lab4*, *Lab5*, *Lab6*

- *Node*: Node number
- *Lab*: UX,UY,UZ,ROTX,ROTY,ROTZ (Translation)
- *Lab2*, *Lab3*, *Lab4*, *Lab5*, *Lab6*: Additional DOF labels

Point Loads , Concentrated LoAD

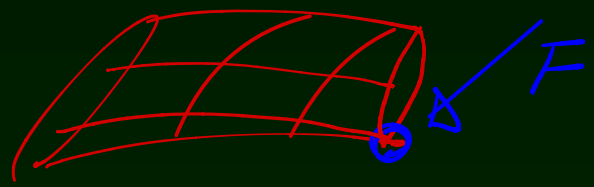
- Forces and moments in a structural analysis
- Heat flow rates in a thermal analysis
- Point loads can be applied to keypoints or nodes.
  (A force on a keypoint will be automatically transferred to the associate node.)

Discipline	Force	ANSYS Label
Structural	Forces Moments	FX, FY, FZ MX, MY, MZ
Thermal	Heat Flow Rate	HEAT, HBOT, HE2, . . . HTOP
Magnetic	Current Segments Magnetic Flux Electrical Charge	CSGX, CSGY, CSGZ FLUX CHRG
Electric	Current Charge	AMPS CHRG
Fluid	Fluid Flow Rate	FLOW

D&F

Point Loads

- Point Loads
  - F : Defines force loads at nodes
  - FK : Defines force loads at keypoints
  - *x*LIST : Lists force loads
  - *x*DELE : Deletes force loads



- Specifies force loads at nodes

F, *Node*, *Lab*, *VALUE*, *VALUE2*, *NEND*, *NINC*

Node number

FX,FY,FZ,MX,MY,MZ

Surface Loads , Pressure.

- Pressures in a structural analysis
- Convections and heat fluxes in a thermal analysis
- Surface loads can be applied to lines or areas (solid model entities), or nodes or elements.
- For surface loads applied to lines or areas, the load is ultimately transferred to individual element faces of the surface.

Surface Loads

lab

Discipline	Surface Load	ANSYS Label
Structural	Pressure	PRES[1]
Thermal	Convection Heat Flux Infinite Surface	CONV HFLUX INF
Magnetic	Maxwell Surface Infinite Surface	MXWF INF
Electric	Maxwell Surface Surface Charge Density Infinite Surface	MXWF CHRGS INF
Fluid	Wall Roughness Fluid-Structure Interface Impedance	FSI IMPD
All	Superelement Load Vector	SELV

Surface Loads

- **Surface Loads**
  - **SF : Defines surface loads on nodes**
  - **SFE : Defines surface loads on elements**
  - **SFL : Specifies surface loads on lines of an area**
  - **SFA : Specifies surface loads on the selected areas**
  - **xLIST : Lists surface loads**
  - **xDELE : Deletes surface loads**

- Specifies surface loads on elements

SFE, *ELEM*, *LKEY*, *Lab*, *KVAL*, *VAL1*, *VAL2*, *VAL3*, *VAL14*

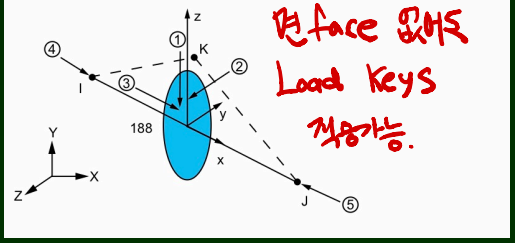
- *ELEM*: Element number
- *LKEY*: Load key
- *KVAL*, *VAL1*, *VAL2*, *VAL3*, *VAL14*: Surface load value

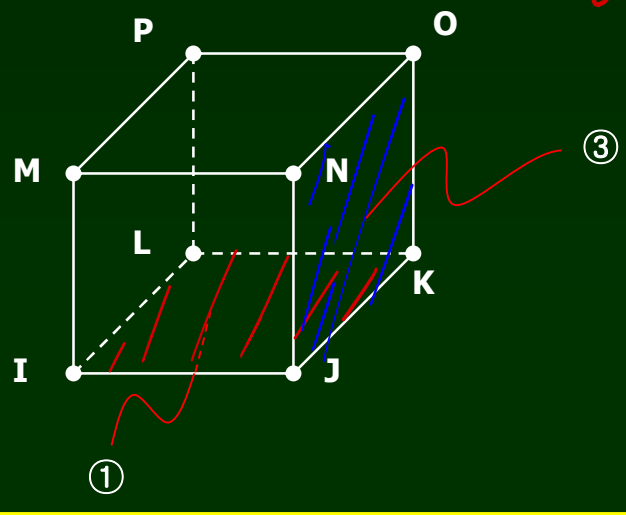
Load Keys

- Load key associated with surface load (defaults to 1). Load keys (1,2,3, etc.) are listed under "Surface Loads" in the input data table for each element type in the *ANSYS Elements Reference*

- For an example,
  Solid185(3-D 8-Node Structural Solid)
  Surface Loads
  Pressures ; face 1 (J-I-L-K), face 2 (I-J-N-M), face 3 (J-K-O-N), face 4 (K-L-P-O), face 5 (L-I-M-P), face 6 (M-N-O-P)

Figure 188.1: BEAM188 Geometry





6개 면 中 1개의 면 선택 (face) ⇩ load keys.

'Which face.?' ⇩ Manual에서 정확 확인.

Body Loads



- Temperatures in a structural analysis
- Heat generations rates in a thermal analysis
- Current densities in a magnetic field analysis
- Body loads can be specified on keypoints or nodes.
  (Body loads on keypoints eventually become groups of body loads on individual nodes.)

Body Loads

Discipline	Body Load	ANSYS Label
Structural	Temperature Fluence	TEMP[1] FLUE
Thermal	Heat Generation Rate	HGEN
Magnetic	Temperature Current Density Virtual Displacement Voltage Drop	TEMP[1] JS MVDI VLTG
Electric	Temperature Volume Charge Density	TEMP[1] CHRGD
Fluid	Heat Generation Rate Force Density	HGEN FORC

Body Loads

Body Loads

- **BF : Defines a nodal body force load**
- **BFE : Defines an element body force load**
- **BFK : Defines a body force load at a keypoint**
- **BFL : Defines a body force load on a line**
- **BFA : Defines a body force load on an area**
- **BFV : Defines a body force load on a volume**
- ***x*LIST : Lists body force loads**
- ***x*DELE : Deletes body force loads**

- Define a nodal body force load

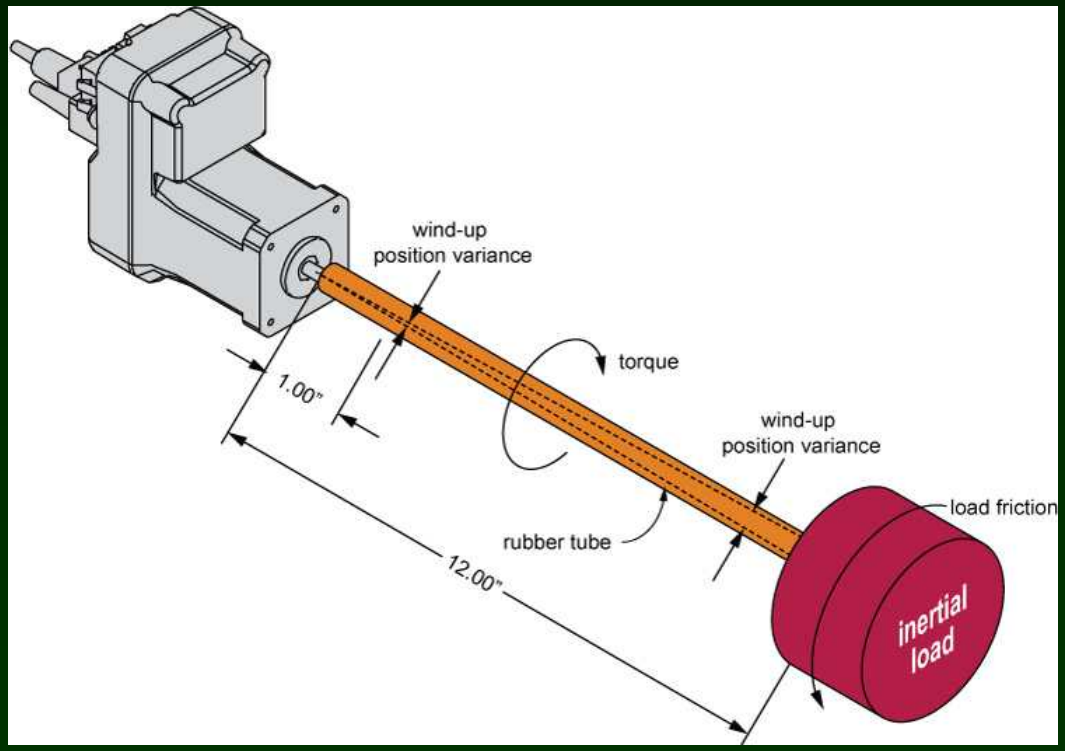
BF, *Node*, *Lab*, *VAL1*, *VAL2*, *VAL3*, *PHASE*

Node number

TEMP, HGEN, etc.

Inertia Loads

- **An inertia load is one that is attribute to the inertia (mass matrix) of a body, such as gravitational acceleration, angular velocity, and angular acceleration.**



Inertia Loads

❑ **Inertia Loads**

- **ACEL : Specifies the linear acceleration of the structure**
- **OMEGA : Specifies the rotational velocity of the structure**
- **DOMEGA : Specifies the rotational acceleration of the structure**

- Specifies the linear acceleration of the structure

ACEL, *ACELX, ACELY, ACELZ*

Linear acceleration of the structure in the global Cartesian X,Y and Z axis directions.

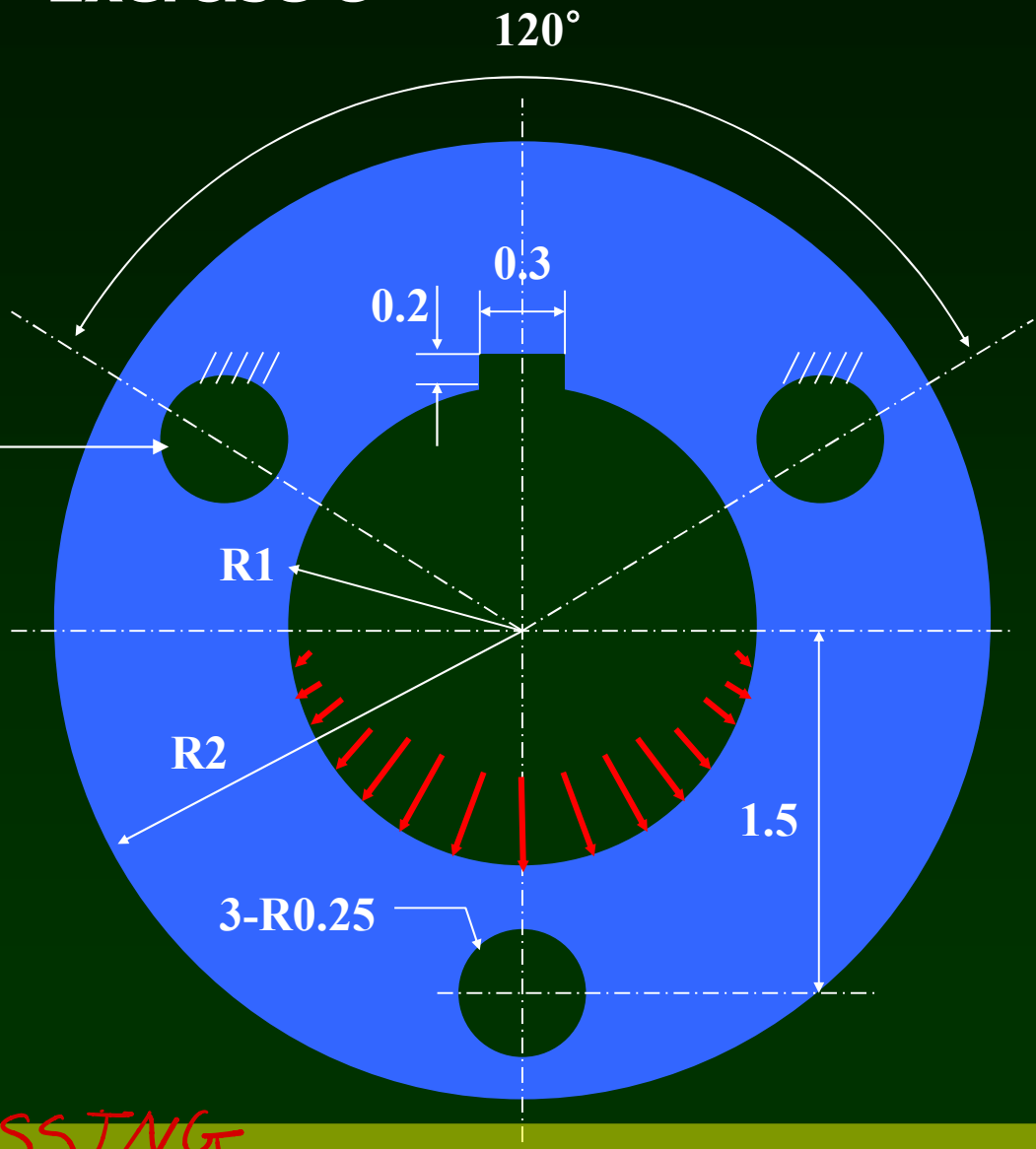
- Specifies the rotational velocity of the structure

OMEGA, *OMEGX, OMEGY, OMEGZ*

Rotational velocity of the structure about the global Cartesian X,Y and Z axes.

Exercise C

SAVE → file.db → RESUME
SOLVE → file.rst



Material properties :

$E = 30 \times 10^6$ lb$_f$/in^2

$\rho = 0.238$ lb/in^3

$\nu = 0.3$

PRE-PROCESSING

SOLVING

Solution

- Enters the solution processor

/SOLU

Solution 진입.

- Specifies the analysis type and restart status

ANTYPE, *Antype, Status, LDSTEP, SUBSTEP, Action*

- Starts a solution command

SOLVE

$[K]\{u\} = [F]$ 풀어.

Solution

- Specifies the analysis type and restart status

ANTYPE, *Antype, Status, LDSTEP, SUBSTEP, Action*

- Analysis type

STATIC or 0 — Perform a static analysis. Valid for all degrees of freedom. , DEFAULT. ANTYPE

BUCKLE or 1 — Perform a buckling analysis. Implies that a previous static solution was performed with prestress effects calculated (PSTRES,ON). Valid for structural degrees of freedom only.

MODAL or 2 — Perform a modal analysis. Valid for structural and fluid degrees of freedom. for Natural Frequency

HARMIC or 3 — Perform a harmonic analysis. Valid for structural, fluid, magnetic, and electrical degrees of freedom. F(w)

TRANS or 4 — Perform a transient analysis. Valid for all degrees of freedom. T

SUBSTR or 7 — Perform a substructure analysis. Valid for all degrees of freedom.

SPECTR or 8 — Perform a spectrum analysis. Implies that a previous modal analysis was performed. Valid for structural degrees of freedom only.

SOIL or 9 — Perform a soil analysis including geostatic stress equilibrium or consolidation. Valid for structural and fluid-pore-pressure degrees of freedom.

Post-Processing

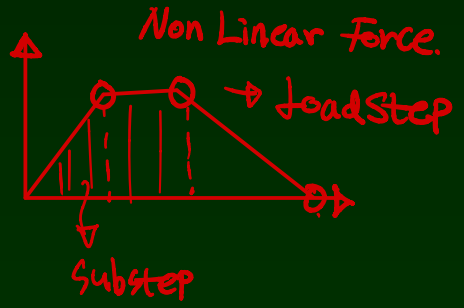
- **Enters the database results postprocessor**

PREP7
POST1 > 차이?
General PRE/POST Process.

/POST1

- **Data base file(*File.DB*)로부터 Model data를 Restore하는 command**

RESUME, *Fname, Ext, Dir, NOPAR*



- **Results file로부터 Result data를 Read하는 command**

결과 READ

Loadstep, SUBstep

SET, *Lstep, SBSTEP, FACT, KIMG, TIME, ANGLE, NSET*

1N, 2N, 3N 있음

Post-Processing

- **Deformed shape displays command**

Plot Displacement.

PLDISP, *KUND*

Key to superimposed undeformed shape:

0 – Deformed shape only (default)

1 – Deformed and undeformed shape

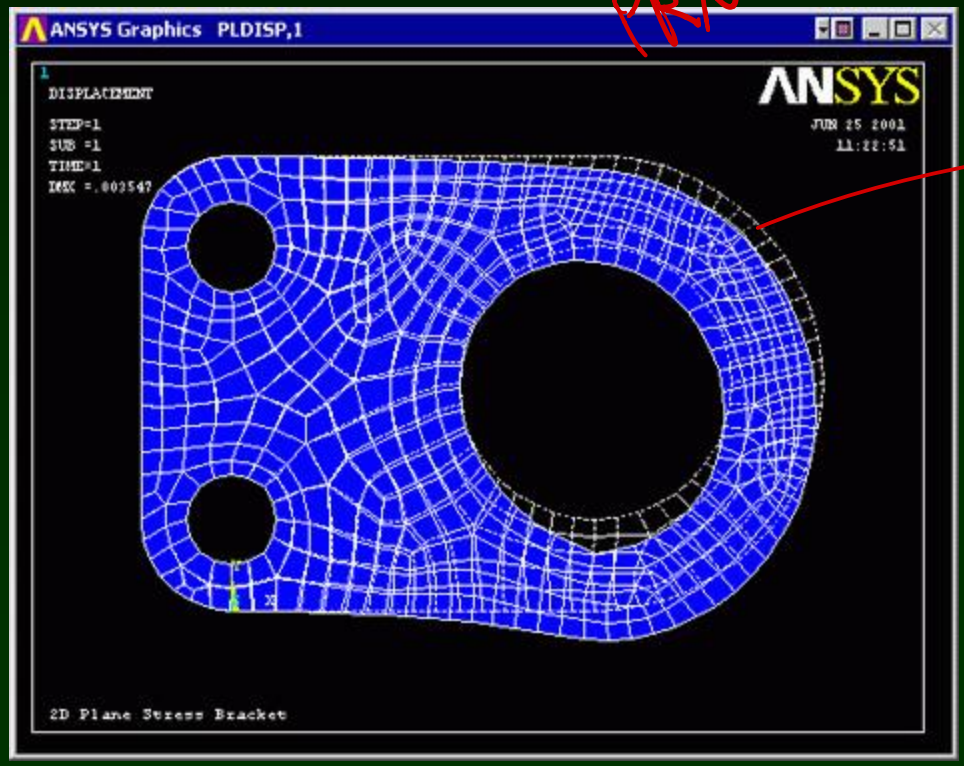
2 – Deformed shape and edges only of undeformed shape

Post-Processing

PLDISP = PLOT Displacement.
PRDISP = Print " "
PLNSOL = plot nodal Sol.
PRNSOL = Print. " "

- **Deformed shape displays command**

PLDISP, 1



Original SHAPE

Post-Processing

- Contour displays command

→ Help에서 "PLNSOL" 검색.

↓

Item & Component 검색.

PLNSOL, *Item*, *Comp*

Component

Item to be contoured:		
S	–	Stress
U	–	Displacements
Temp	–	Temperature
etc.		
Vector items also require a component		

Component of *Item* (if required):		
X, Y, Z	–	Stress and displacement components
XY, YZ, XZ	–	Shear stress components
1, 2, 3	–	Principal stresses
INT	–	Stress intensity
EQV	–	Equivalent stress

If σ_x 가 궁금,

PLNSOL, S, X

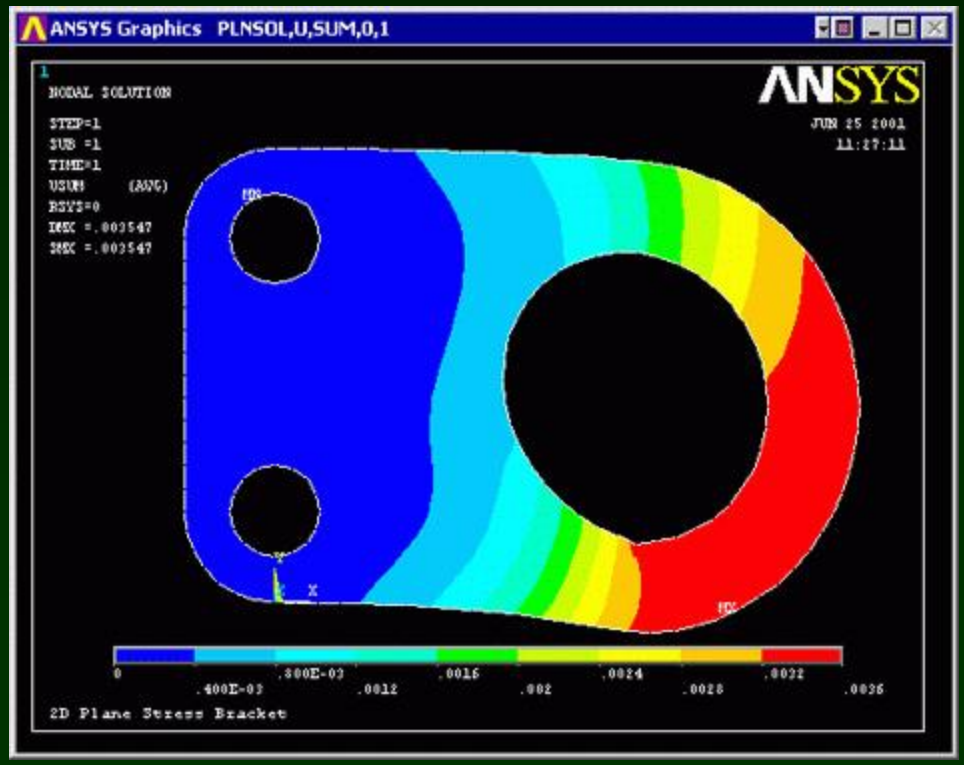
↓

stress

Post-Processing

- **Contour displays command**

PLNSOL, U, SUM



ANSYS Parametric Design Language(APDL)

- **Defining Parameters**

```
*SET,ABC,-24
*SET,QR,2.07E11
*SET,XORY,ABC
*SET,CPARM,'CASE1'
```

```
ABC=-24
QR=2.07E11
XORY=ABC
CPARM='CASE1'
```

- **Assigning Mechanical APDL-Provided Values to Parameters(Using *GET command)**

*GET,*Par,Entity,ENTNUM,Item1,IT1NUM,Item2,IT2NUM*

*GET,BCD,ELEM,97,ATTR,MAT	BCD = Material number of element 97
*GET,V37,ELEM,37,VOLU	V37 = Volume of element 37
*GET,NX52,NODE,52,LOC,X	NX52 = x location in the active coordinate system of node 52
*GET,NMAX,NODE,,NUM,MAX	NMAX = Highest node number in the selected set
*GET,NCOUNT,NODE,,COUNT	NCOUNT = Number of nodes in the selected set
*GET,UX12,NODE,12,U,X	UX12 = X structural displacement of node 12
*GET,ETAB28,ELEM,28,ETAB,*Lab*	ETAB28 = Value of element table item *LAB* for element 28
*GET,COORD,ACTIVE,,CSYS	COORD = Active coordinate system number

ANSYS Parametric Design Language(APDL)

- **Assigning Mechanical APDL-Provided Values to Parameters (Using Get Function)**

```
*GET,L1,NODE,1,LOC,X
```

```
L1=NX(1)
```

NX(*N*)	X-coordinate of node *N* in the active coordinate system.
KX(*K*)	X-coordinate of keypoint *K* in the active coordinate system
NODE(*X,Y,Z*)	Number of the selected node nearest the *X,Y,Z* point
KP(*X,Y,Z*)	Number of the selected keypoint nearest the *X,Y,Z* point
DISTND(*N1,N2*)	Distance between nodes *N1* and *N2*.
DISTKP(*K1,K2*)	Distance between keypoints *K1* and *K2*.
NDNEXT(*N*)	Next selected node having a node number greater than *N*.
ELNEXT(*E*)	Next selected node having an element number greater than *E*.
ENEXTN(*N,LOC*)	Element connected to node *N*. *LOC* is the position in the resulting list when many elements share the node
NELEM(*E,NPOS*)	Node number in position NPOS (1--20) of element *E*.
UX(*N*)	UX structural displacement at node *N*.

ANSYS Parametric Design Language(APDL)

- Process Control

Commands	Descriptions
*DO	Defines the beginning of a do-loop.
*DOWHILE	Loops repeatedly through the next *ENDDO command.
*ELSE	Separates the final if-then-else block.
*ELSEIF	Separates an intermediate if-then-else block.
*ENDDO	Ends a do-loop and starts the looping action.
*ENDIF	Ends an if-then-else.
*EXIT	Exits a do-loop.
*GO	Causes a specified line on the input file to be read next.
*IF	Conditionally causes commands to be read.

ANSYS Parametric Design Language(APDL)

- **Parametric Expression**

Operator	Operation
+	Add
-	Subtraction
*	Multiplication
/	Division
**	Exponentiation
<	Less-Than Comparison
>	Greater-Than Comparison

```
X=A+B
P=(R2+R1)/2
D=-B+(E**2)-(4*A*C)
```

$$D = -B + E^2 - 4AC$$

```
XYZ=(A<B)+Y**2
```

$$XYZ = A + Y^2 \quad if \ A < B$$
$$XYZ = B + Y^2 \quad if \ A \geq B$$

```
INC=A1+(31.4/9)
M=((X2-X1)**2-(Y2-Y1)**2)/2
```

ANSYS Parametric Design Language(APDL)

- Parametric Functions

Functions	Descriptions
ABS(x)	Absolute value of x
EXP(x)	Exponential of x : e^x
LOG(x)	Natural log of x : ln (x)
LOG10(x)	Common log of x : $\log_{10}(x)$
SQRT(x)	Square root of x
SIN(x),COS(x), TAN(x)	Sine, Cosine, and Tangent of x. x is in radians by default.
ASIN(x),ACOS(x), ATAN(x)	Arcsine, Arccosine, and Arctangent of x. x must be between -1.0 and +1.0 for ASIN and ACOS. Range of output is $-\pi/2$ to $+\pi/2$ for ASIN and ATAN, and 0 to π for ACOS in radians by default.
MOD(x,y)	Remainder of x/y, computed as x - (NINT(x/y) * y). y=0 returns zero (0).